

**EVALUATING MOTION SICKNESS FACTORS IN VR: THE ROLE OF
NAVIGATION, INTERACTION AND ENVIRONMENTAL EFFECTS**

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ABSTRACT

Interacting with virtual reality (VR) presents a persistent challenge: motion sickness, particularly during immersive VR game play. Discomfort is influenced by hardware specifications, user characteristics, and individual susceptibility. Additionally, game mechanics, interaction modes, and environmental design significantly impact user experience and can exacerbate VR-induced sickness.

To investigate the root causes of VR motion sickness, this study utilizes two custom-designed gaming prototypes built in Unity’s Universal Render Pipeline (URP). These prototypes feature a sci-fi outer space environment with zero-gravity simulation. One version includes basic sci-fi assets and movement mechanics, while the other incorporates physics-based rendering (PBR), visual effects (FX), and timed interactive tasks. Both are tested using Meta Quest 3 and Quest 3s headsets.

Swarm VR, a commercial VR title, is included as a benchmark to assess baseline user responses. Each participant engages in three game play phases totaling approximately 30 minutes: 7 minutes with Swarm VR, 7 minutes with the basic IGR prototype, and 16 minutes with the advanced IGR game play experience. Participants complete the Motion Sickness Questionnaire (MSQ) beforehand and provide feedback through the Virtual Reality Sickness Questionnaire (VRSQ) after each session.

The experiment is conducted with 33 participants, each progressing through game play scenarios of increasing difficulty. Their feedback is used to analyze visual and interaction-based factors contributing to VR sickness. Results derived from MSQ and VRSQ responses help identify high-risk elements, contributing to design guidelines for minimizing discomfort and improving immersive VR interaction quality.

Index Terms-Virtual Reality (VR), VR Sickness, Motion Sickness, Head-Mounted Dis-

plays (HMDs), VR Locomotion, jet packing, Web-Swinging, Immersive Experience, game play Adaptation, Training Modules, Story-Driven Design, SSQ, VRSQ, MSQ, Oculomotor Strain, Disorientation, HRV, EDA, Meta Quest 3, Adaptive game play, Swarm VR Benchmark, Field of View (FoV), Real-Time Interaction, Latency, Spatial Orientation.

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Chapter 1

Introduction

Virtual reality (VR) has become one of the most innovative technologies with tremendous potential to transform various industries. VR is increasingly being used in fields such as healthcare, where it helps in medical training and rehabilitation [10], gaming, providing immersive experiences [6], and education, offering virtual learning environments [20]. Additionally, VR has found applications in architecture for virtual walkthroughs [28], and in the aerospace sector for simulating flight conditions [18]. It offers users immersive experiences that replicate both real-world settings and imaginative, fantastical environments. Even with the tremendous research in this area, the technology continues to evolve rapidly, with many recent developments focused on improving realism and user engagement in virtual spaces [10][28].

A visual representation of VR applications across these industries is shown in Figure 1.1.

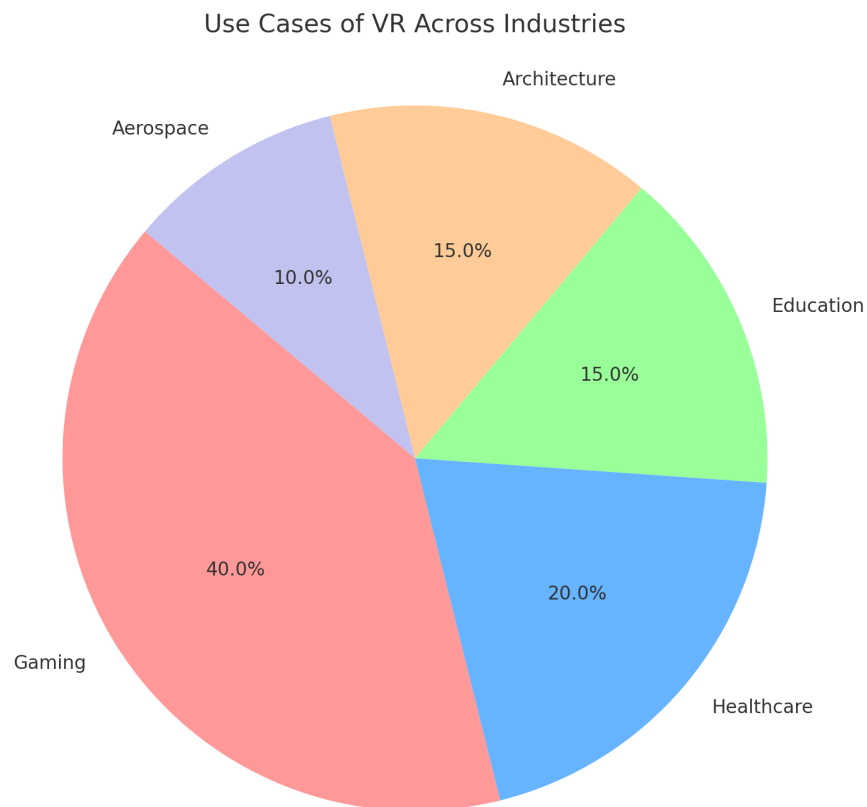


Figure 1.1: Distribution of Virtual Reality Applications Across Major Industries. Based on sources from Kim et al. (2018) [20], Haamer et al. (2024) [10], and Webb et al. (2021) [28].

However, the immersive quality of VR that drives user engagement may paradoxically contribute to motion-related discomfort. Presence and cybersickness are often inversely related; higher presence can sometimes alleviate symptoms, while in other cases it may intensify them [29]. The level of immersion and the type of navigational task being performed significantly influence symptom onset. For instance, users engaging in high-immersion navigation tasks tend to experience greater discomfort than those in passive VR environments [8]. Virtual reality's greatest challenge is motion sickness, also referred to as VR sickness or cybersickness. It remains a significant barrier to mainstream VR adoption, particularly in scenarios involving prolonged or complex interaction [12] [11]. It occurs when there is

inconsistency in the sensory input, primarily between what the users experience through the virtual environment and their inner ear sense of movement. It leads to symptoms of nausea, dizziness, oculomotor distress, and disorientation [18] [20]. Motion sickness is a negation of user experience and hinders technological uptake.

Although VR motion sickness is already a common problem, it should be mentioned that different software tests different facets of the illness. For example, the VRSQ (Virtual Reality Sickness Questionnaire) of Kim et al. [20] is better as it measures only symptoms most relevant to modern head-mounted displays (e.g., eye fatigue and disorientation), while the SSQ (Simulator Sickness Questionnaire) of Kennedy et al. [18] is an outdated measure that assesses general symptoms such as nausea and oculomotor fatigue. On the other hand, the MSQ will be most useful in establishing a participant’s baseline because it quantifies general motion sickness susceptibility without regard to VR parameters. The symptoms that each tool measures are compiled in Table 1.1.

Symptom Type	SSQ [18]	VRSQ [20]	MSQ [18]
Oculomotor Strain	✓	✓	—
Nausea	✓	—	✓
General Susceptibility	—	—	✓

Table 1.1: Comparison of VR sickness symptoms tracked by SSQ, VRSQ, and MSQ.

Recent research has focused on reducing VR motion sickness using a number of methods, including maximizing visual fidelity, motion congruency, and responsiveness of hardware. For example, it is stated by studies that reducing motion-to-photon latency, adjusting field of view (FoV), and limiting peripheral motion effects will reduce symptoms [6][20].

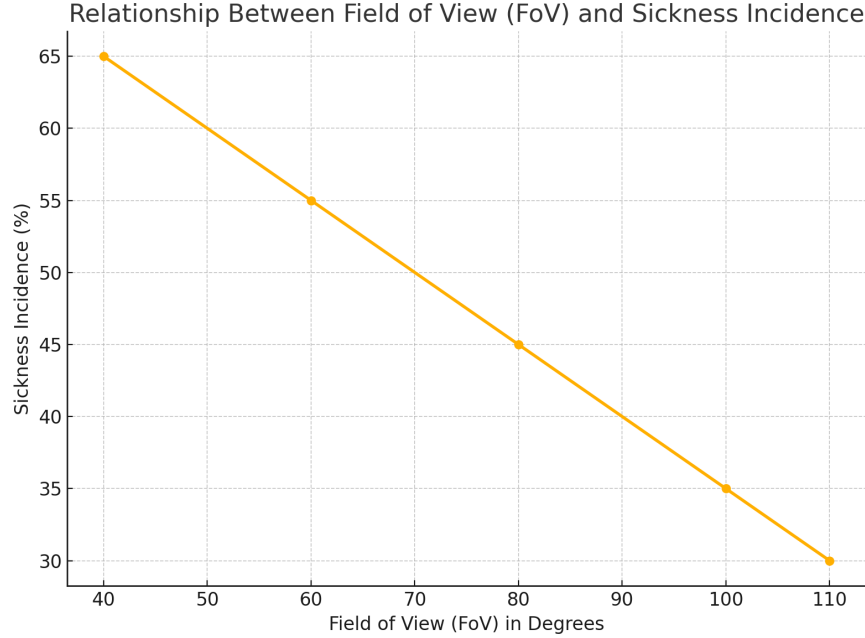


Figure 1.2: Relationship Between Field of View (FoV) and Sickness Incidence in VR. As FoV increases, sickness symptoms decrease, highlighting a trade-off between comfort and immersion.[28][6]

The graph above 1.2 shows the relationship between field of view (FoV) and the prevalence of VR sickness. It shows the trend described by Webb et al. (2021) and Chang et al. (2020): with increasing FoV, incidence of motion sickness reduces-though higher FoVs bring other immersion or hardware trade-offs.

Causes of VR sickness include hardware parameters (latency, refresh rate, and tracking accuracy), user attributes (age, gender, and motion sickness susceptibility), and virtual environment characteristics (movement dynamics, scene complexity, and camera movement) [25][28]. Discomfort intensity typically arises from the complex interdependence of these factors.

Designing naturalistic experiences and naturalistic interactions are significant factors in the immersive experience of VR.

It often includes achieving a compromise between motion mechanics, graphical fidelity, and system performance. In the case of motion sickness, factors like field of view (FoV), camera movement, latency, post-processing effects, and individual sensitivities (e.g., age, gender, prior VR experience) play a significant role [10][20]. In addition all the other causes, individual factors such as stress levels, anxiety, smoking history, and susceptibility to motion sickness also play a crucial role in VR discomfort [19].

In recent times, research identifies the locomotion system as one of the most crucial components influencing user comfort in VR; the selection of navigation techniques such as teleportation, smooth walking, gesture-based movement, free locomotion, web swinging, and jet packing can significantly affect VR sickness and user engagement [5] [10] [28]. Existing research identifies that decreased FoV values will reduce motion sickness, but this is generally at the cost of immersion and user engagement [28][6]. Furthermore, individual differences such as susceptibility to visually induced motion sickness highlight the value of adaptive VR experiences [20][25]. This widespread concern means finding a balance between reducing discomfort and maximizing the immersive nature that defines VR [28][6].enhancing the immersive nature that defines VR [28][6].

1.0.1 Background

Crafting realistic experiences and intuitive interactions are key components in the immersive nature of VR. It often requires a balance between motion mechanics, graphical fidelity, and system performance. In the onset of motion sickness, features like field of view (FoV), camera motion, latency, post-processing effects, and individual susceptibilities (e.g., age, gender, prior VR experience) play a significant role [10][20]. Mechanics like teleportation, free locomotion, and jet packing for movement have significant influence on user comfort [10][28]. Past research emphasizes that the smaller FoV values can reduce motion sickness, which often compromises user immersion and engagement [10][28]. Moreover, individual

differences, such as susceptibility to visually induced motion sickness, highlight the need for adaptive VR experiences [20][28]. It requires recognizing a balance between minimizing discomfort and enhancing the immersive nature that defines VR [28][6].

1.0.2 Research Objective

Addressing the challenges of motion sickness and setting guidelines for designing comfortable virtual reality (VR) experiences through a controlled experimental approach is the aim of this study. The research uses two in-house-developed VR testbeds and one comparative VR experience to serve as the platforms in the research for testing the causative factors of motion sickness and identifying factors that lead to user comfort and immersion. The primary research objectives are outlined below.

1. Identification of the Key Factors Contributing to VR-Induced Motion Sickness:

VR motion sickness is caused by different factors that are interrelated in nature, i.e., hardware performance, rendering techniques, user susceptibility, and virtual environment characteristics. Motion-to-photon latency, field of view (FoV), peripheral motion, and camera movement inconsistency influence user comfort [18][6][10]. This study uses a pre-experimental Motion Sickness Susceptibility Questionnaire (MSQ) to quantify participants' baseline tolerance and identifies key discomfort contributors by utilizing post-experience feedback and the Virtual Reality Sickness Questionnaire (VRSQ) [20]. By comparing participant response during three distinct game play sessions Swarm VR, a low-fidelity IGR prototype, and a high-fidelity FX IGR prototype this research aims to find factors such as asset fidelity, FX, environment design, navigation mode, and interaction complexity that contribute to VR-induced motion sickness [25][28].

2. Evaluation of User Comfort and Immersion using a Variety of game play

Mechanisms in a Zero-Gravity Sci-Fi Environment:

The second objective is to quantify the effect of different interaction and navigation mechanics on user comfort and immersion in VR. The two IGR prototypes differ along visual and interaction fidelity: the first has high-end FX and physics-based rendering (PBR), whereas the second uses slimmer sci-fi objects with low-impact interactions. The game mechanic includes movement mechanics such as jet packing and web-swinging within a zero-gravity outer space environment. Empirical data show that users' belief in natural and congruent movement influences sickness and immersion degrees significantly [10][28][24]. Furthermore, existing research points out that locomotion is one of the most significant factors in user comfort in VR; navigation method selection, e.g., teleportation, smooth walking, or gesture-based motion, can impact both motion sickness and engagement significantly [5]. By comparing user feedback for all three experiences, this study will assess the relationship between interaction style, task difficulty, and comfort. There is a VRSQ following each session to measure the degree of discomfort and identify which mechanics or designs enhance or degrade immersion [20][25].

3. Developing Guidelines for Designing VR Experiences That Reduce Discomfort While Immersion and Engagement Are Maintained:

Recent studies are exploring vestibular stimulation as a promising method to reduce cybersickness and increase user comfort during extended VR use [27]. This objective attempts to establish evidence-based VR design guidelines for VR developers to provide reduced motion sickness while ensuring immersive quality is maintained. From all three experiences offered in this study, the aim is to recommend the best balance between visual complexity, interaction mechanism, locomotion approach, and system latency that supports varying user tolerances. Past research points out that the match between

visual and vestibular signals, along with latency and FoV settings that are adjusted to reduce motion sickness without impacting engagement [18][6][28]. According to Casas et al. [5], the correct locomotion approach is essential to fulfill both comfort and presence within virtual environments. This study contributes to best practices in FX, navigation mechanics, and environmental integration that ensure presence and comfort in VR worlds.

1.1 Prototype Overview

In order to understand how the effects of different locomotion systems, interaction complexities, and environmental factors influence VR-induced motion sickness and comfort of users in virtual reality (VR) experiences, two immersive experimental prototypes were developed using Unity’s Universal Render Pipeline (URP). Each is mounted on Meta Quest 3 and 3s headsets and simulates a zero-gravity outer space environment typical of sci-fi environments, with floating debris around and a core in the center. These prototypes are built on the basis of existing literature on the influence of cybersickness and the usability of VR [28][5][10].

Prototype 1: Bare Bones VR Experience The first prototype serves as a control, with a conscious reduction in scope and complexity to isolate the effect of locomotion and basic interaction from visual or cognitive distraction.

1. **Navigation:** This prototype includes two locomotion systems:

- (a) *Web-Swinging*: Engages with floating asteroids and fires from a grip-based anchor-and-release system to mimic grappling-like motion. Anchor-point travel provides continuous locomotion, which has been shown to increase the likelihood of disorientation in some players [5].
- (b) *jet packing*: Characters are propelled forward in the direction of controller orien-

tation while gripping, allowing for constant propulsion and smooth movement. It was found through research that constant, repeatable motion systems such as jet packing have been found to provide lower cybersickness levels [28] [10].

2. **Environment:** The prototype features basic sci-fi assets, low-poly modular debris, and plain lighting. No dynamic shaders, VFX, or ambient background sounds exist. This basic environment allows sickness responses to be easily a product of the mechanics and not environmental complexity.
3. **Interaction:** The players operate a grip control to capture and throw space trash into a point energy core. The task is timed to simulate rudimentary user interaction under conditions of low pressure.
4. **Objective:** Evaluate user comfort and motion sickness induced by simple locomotion techniques within a sparse visual environment. Existing research by Kim et al. [20] and Mazloui et al. [25] identifies the necessity of isolating visual stimuli to accurately quantify physiological discomfort.

Prototype 2: Feature-Rich VR Experience

This prototype is an enhanced version of the first prototype in terms of environmental richness, added game play, navigation and interaction controls, and structured user onboarding. This game play will be utilized to comprehend the influence of cognitive load, environmental fidelity, and multitasking on VR-induced motion sickness, presence, and user investment.

1. **Navigation:** Like IGR Old Prototype, this version includes movement mechanics like Web-Swinging: grip-based traversal of floating asteroids and jet packing (smooth directional thrust managed by grip and direction of orientation). This setup enables

us to compare directly with the minimum prototype to assess sickness variation when there is sensory stimulation.

2. **Environment:** A deep space sci-fi environment is created through the use of PBR (Physically-Based Rendering) materials, orbiting space debris, dynamic post-process effects (i.e., bloom, chromatic aberration), and space ambient audio. Since research by Kim et al. [20] and Haamer et al. [10] verifies that visual stimulation intensity and complexity are the chief causative factors for VR sickness, this prototype is ideal for threshold shift testing.
3. **Interaction and Task Complexity:** Apart from picking up and tossing debris into energy cores, players now have to deal with dynamic threats and fight against enemy drones. Players use a gun fueled by the trigger button to shoot enemies coming without colliding with jet pack movement. These interactions provide an approximation of realistic multitasking under stress, which is known to impact sickness tolerance and task performance [25][20].
4. **Training Modules:**
 - (a) **Tutorial 1 Web-Swinging and Space Debris Collection:** A short video and interactive tutorial introduce users to anchor-based movement and grip-throw mechanics.
 - (b) **Tutorial 2 jet packing, Space Debris Collection, and Combat:** A specific training space enables learners to acquire thrust navigation and gunplay control.

Tutorials in sequence enable participants to get accustomed to the control systems prior to being exposed to high-intensity stimuli. Phased onboarding facilitates ongoing work that asserts that familiarization diminishes sensory mismatch and VR-induced sickness [10][24].

5. **Narrative and Immersive Design:** Players are assigned the role of an *Intergalactic Ranger* tasked with clearing space debris and protecting important energy sources. The introduction of voiceover, the goal-oriented HUD, and reactive ambient objects define mission conditions. Haamer et al. [10] and Lonner et al. [24] hypothesize that a goal-oriented story helps to maintain user focus, which can reduce disorientation through cognitive anchoring. IGR Old Prototype provides a low, distraction-free environment in which to pilot baseline sickness factors for locomotion and simple object manipulation. IGR New game play includes high-fidelity graphics, multitasking, onboarding guidance, and combat features intended to test the tolerance limits of the user within a test environment. Compare participant reaction across the two prototypes; this study will identify significant experiential and technical parameters governing motion sickness in VR.

Chapter 2

Literature Review

Motion sickness remains one of the principal constraints on the prolonged use of immersive VR systems, particularly in dynamic environments with difficult locomotion and interaction tasks. Research in neurophysiology also links motion sickness to vestibular conflicts and brain integration failures [17][31]. The literature attributes VR-induced sickness to multifactorial causes, including hardware limitations, user-specific characteristics, visual complexity, and cognitive load. Increased realism, wider field of view (FoV), and longer sessions can increase the chances of experiencing cybersickness. This is particularly evident when wide FoV is paired with complex motion dynamics or scene transitions, leading to sensory overload [26][9].

This review gathers relevant research to explore these contributors and search for how this research bridges some of the most significant gaps in the research.

2.0.1 VR Sickness: Measurement and Definitions

The Simulator Sickness Questionnaire (SSQ) from Kennedy et al. [18] is still among the most commonly applied tools for assessing simulator discomfort. It distinguishes between nausea, oculomotor disturbance, and disorientation. Nonetheless, its suitability for current head-mounted VR systems is restricted because no HMD-specific variables are included.

To address these limitations, Kim et al. [20] developed the Virtual Reality Sickness Questionnaire (VRSQ) that omits nausea and focuses on disorientation and oculomotor stress. It is especially suited to short, interactive VR experiences, such as those employed in consumer HMDs like the Meta Quest 3.

Additionally, the Cybersickness in Virtual Reality Questionnaire (CSQ-VR) introduced by Kourtesis et al. offers an alternative VR-specific metric that incorporates both physical and perceptual discomfort while improving ecological validity in modern immersive systems [21].

Recent studies also explore the influence of vestibular stimulation on discomfort perception, suggesting that feedback mechanisms targeting inner ear response may reduce VR-induced nausea [27].

2.0.2 Contributing Factors to Motion Sickness

Chang et al. [6] divided VR sickness factors into three categories: hardware, content, and user factors. This classification, confirmed by Webb et al. [28], identifies a number of high-risk variables:

1. **Hardware Latency and Field of View (FoV):** Motion-to-photon latency delays above 20 ms are found to lead to an increase in discomfort. Low FoVs reduce sickness but perhaps at the cost of immersion [6][28].
2. **Visual Design:** Abrupt camera motion, over-post-processing (bloom, blur, distortion), and visual ungrounding increase sickness risk [28].
3. **Vection and Visual Flow:** Passive self-motion illusions (vection) induced by visual stimuli contribute significantly to motion sickness in VR, particularly in driving and floating experiences [13].

4. **Post Processing and Environment settings:** Foveated rendering, dynamic blur, and adaptive VR environments tailored to user profiles [15] [23]
5. **User-Specific Factors:** Demographics and personal traits play a role in susceptibility. Female users
6. **User-Specific Factors:** Demographics and personal traits play a role in susceptibility. Female users have been found to be more uncomfortable than males, and VR experience alleviates symptoms through adaptation [20][28][3]. Weech et al. [30] found that motion sensitivity, vestibular function, and anxiety are key individual predictors of cybersickness severity. Weech et al. [30] found that motion sensitivity, vestibular function, and anxiety are key individual predictors of cybersickness severity. [20][28].
7. **Cognitive Load and Stress Response:** Real-time physiological markers such as heart rate and reaction time are increasingly used to evaluate cognitive overload, which correlates with increased motion discomfort [22].
8. **Sensorimotor Mismatch and Age:** Conflicting sensory signals (especially between vestibular and visual systems) cause disorientation. These effects are more pronounced in older users due to slower sensory adaptation [16].

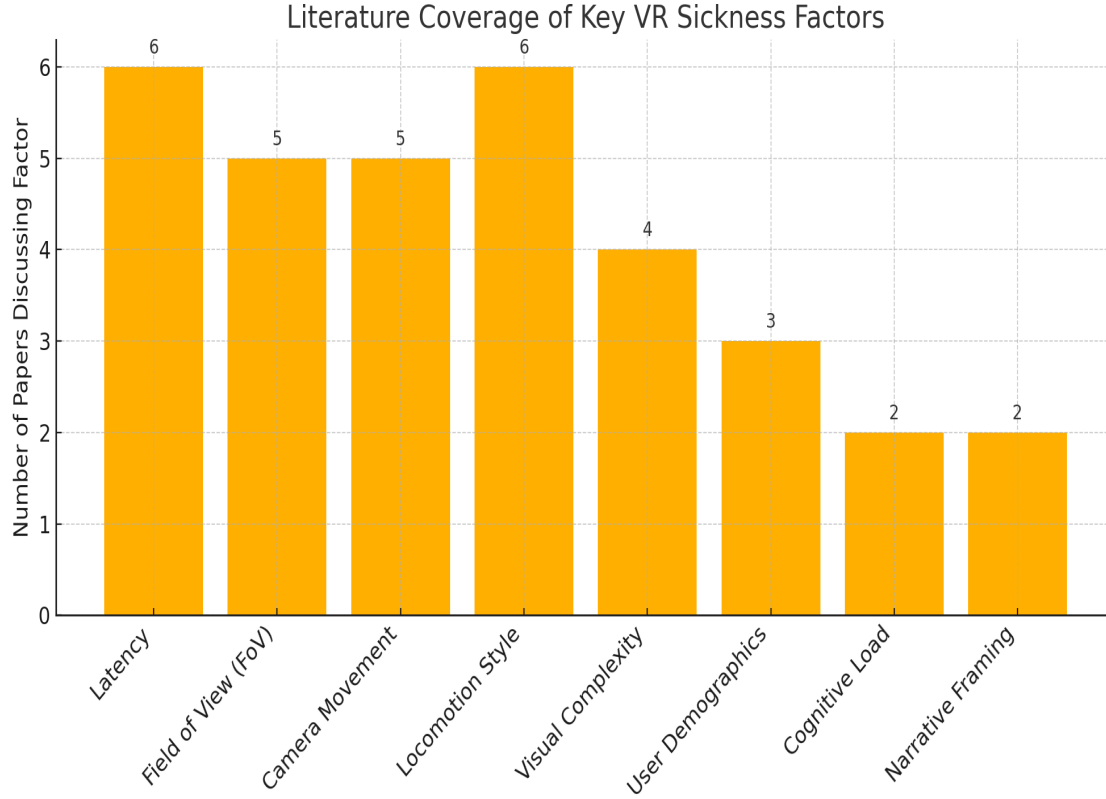


Figure 2.1: Frequency of key VR motion sickness factors covered across recent literature.

A review of prominent VR sickness research reveals an uneven distribution of attention across contributing factors. As shown in Figure 2.1, core technical issues such as latency, field of view (FoV), and locomotion style are consistently explored. However, cognitive load and narrative framing-elements increasingly relevant in immersive VR applications remain comparatively underrepresented. This highlights the need for a more holistic approach, as investigated in this study [6][28][10][5][20][24].

These findings point to the requirement for a lot of testing of these variables within real-world VR settings, rather than controlled laboratory situations.

Table 2.1: Key findings and mitigation strategies for VR motion sickness.

Factor	Contributing Element	Proposed Mitigation	References
Field of View (FoV)	Narrow FoV reduces sickness but limits immersion	Use dynamic FoV adjustments based on task intensity	[6][28]
Latency	High motion-to-photon latency causes nausea	Maintain latency under 20ms for real-time feedback	[18][6]
Camera Movement	Rapid or non-user-controlled motion increases discomfort	Implement user-driven or gradual camera transitions	[20][28]
Locomotion Style	Abrupt or unnatural movement (e.g., web-swinging) increases sickness	Favor continuous, smooth navigation like jet packing	[5][10]
Visual Complexity	Excessive post-processing, motion blur, cluttered visuals	Use moderate visual effects; reduce unnecessary stimuli	[6][28]
Peripheral Vision Effects	Visual conflict in peripheral field causes disorientation	Apply black vignettes or peripheral blur during motion	[28][10]
Continued on next page			

Table 2.1 – continued from previous page

Factor	Contributing Element	Proposed Mitigation	References
Interaction Mechanics	Complex interactions (e.g., throwing, shooting) increase cognitive load	Introduce through tutorial modules and progressive learning	[20][10]
User Demographics	Age, gender, prior VR experience affect tolerance	Tailor game play intensity and offer accessibility options	[20][28]
Narrative Engagement	Lack of contextual story increases perceived disorientation	Embed goal-driven storylines to maintain focus and reduce sickness	[24] [10]
Onboarding/Training	Immediate exposure to complex tasks causes sensory overload	Use structured training modules with gradual difficulty increase	[10][28]
Prototype Realism	Simple test environments don't reflect real usage scenarios	Build multi-mechanic prototypes with progressive challenges	[5][10][28]
Evaluation Metrics	Traditional SSQ lacks specificity for VR interaction	Use VR-specific tools like VRSQ and MSQ	[20][10]
Continued on next page			

Table 2.1 – continued from previous page

Factor	Contributing Element	Proposed Mitigation	References
Sensorimotor Mismatch	Conflict between expected and perceived movement	Use motion cues aligned with vestibular expectations, especially for older users	[16]
Visual Flow /vection	Illusory self-motion via strong optic flow	Introduce scene-stabilizing cues or reduce peripheral motion	[13]
Vestibular Feedback	Poor vestibular alignment causes nausea	Integrate vestibular stimulation or feedback alignment tools	[27]
Physiological Metrics	Heart rate and accuracy drop under stress	Monitor physiological load to balance difficulty in real-time	[22]

Locomotion Systems and Their Impacts

Locomotion is of the highest importance in VR sickness. Casas et al. [5] surveyed locomotion types and determined that continuous and smooth systems such as jet packing feel more comfortable than abrupt or momentum systems such as web-swinging. Such results have been corroborated by Haamer et al. [10] and Webb et al. [28], who found that motion

models aligned to real-life expectations reduce vestibular-visual mismatch [4]. Prior work by Borst and Volz demonstrates that constrained, treadmill-based locomotion interfaces can reducevection-induced sickness compared to free movement, providing a physically grounded sense of motion [4].

Locomotion Type	Sickness Risk	Immersion Level
Teleportation	Low	Medium
Smooth Walking	Medium	High
jet packing	Medium-Low	High
Web-Swinging	High	High
Gesture Navigation	Medium	High
Treadmill-Based	Low	Medium-High

Table 2.2: Comparative analysis of VR locomotion styles.

jet packing offers smooth forward progression, while web-swinging introduces high angular velocity and trajectory uncertainty, with a higher likelihood of discomfort. This study isolates and tests them separately to determine their individual and cumulative effect. It is evident that the locomotion style dramatically influences user comfort. The following risk matrix 2.2, synthesized from the literature, helps frame the comparative risks of each navigation style.

Cognitive Load and Interaction Complexity

Mechanics of interaction support user immersion but impose cognitive and perceptual burdens. Actions such as grasping, tossing, and shooting, especially under time pressures, can increase sensory overload if imposed suddenly.

Kim et al. [20] and Webb et al. [28] both show that natural, intuitive interactions will

alleviate discomfort when augmented with guided onboarding. Haamer et al. [10] showed that training modules significantly increase user adaptation, minimizing negative symptoms when playing.

Narrative Framing and Environmental Design

Immersive storytelling can also serve as a psychological countermeasure to VR sickness [29]. Lonner et al. [24] experimented by submerging participants in a story-driven mission and found that they showed fewer symptoms of sickness due to greater concentration and mental presence. Haamer et al. [10] also found that realism of the environment, as well as narrative framework and scripted mission briefings, resulted in decreased disorientation and enhanced user comfort.

This is the reason why voiceover briefings, HUD objectives, and ambient noise have been included in IGR New game play of this study.

2.0.3 Limitations of Research in the Literature

Despite advancements, a number of areas of limitation persist with existing VR sickness research:

1. **Lack of Integrated Analysis:** Most studies work on individual variables such as latency or locomotion mode without checking combined effects within interactive scenarios [6][20]. Recent work emphasizes the need for immersive studies integrating vection, vestibular feedback, cognitive stress, and environmental design into a unified testbed [27][13].
2. **Simplicity of the Prototype:** Previous research tends to have simplified test conditions without visuals, enemies, or game play loops, limiting ecological validity [28][10].

3. **Under-utilization of VR-Specific Measures:** Though SSQ is common, VRSQ is not utilized in measuring rich, multi-phase prototypes of real VR usage [20]. Few studies apply real-time physiological metrics in VR discomfort measurement, despite their increasing relevance in simulation-based evaluation [22].
4. **No Progressive Difficulty:** True VR games have task complexity increase and multitasking. Current experiments do not have this progression, constraining insights into long-term adaptation [10].
5. **Limited Narrative Evaluation:** There are not many studies on how narrative structure moderates motion sickness by varying attentional focus and engagement [24].
6. **Text Legibility and Visual Communication:** In scenarios where textual overlays or mission briefings are used, foggy or complex environments reduce legibility, leading to missed cues and user confusion [14].

2.0.4 Study Contributions

The study addresses these issues through the development of two prototypes of VR:

1. **Prototype 1:** Low-fidelity setup where only web-swinging, timed object-throwing, and jet packing are included. There is no training, visual effects, or enemies.
2. **Prototype 2:** High-fidelity, fully immersive version that includes progressive tutorials, real audio-visual effects, jet pack movement, web-swinging, grabbing, shooting, and an overarching narrative where the user is an "Intergalactic Ranger."

2.0.5 Gaps and Limitations in Current Research

While much research has investigated the causes and effects of VR motion sickness, previous work presents a number of recurring limitations that restrict validity to actual VR appli-

cation. These include narrow holistic investigation, low-fidelity prototyping, demographic homogeneity, stationary environments, and absence of measurement tools. All these limitations are resolved in this study by developing interactive prototypes with high-intensity locomotion, narrative immersion, adaptive challenge, and rich user feedback mechanisms.

Lack of Multi-Factor Integration

Most prior work separates individual factors like latency, field of view (FoV), or specific navigation techniques (e.g., teleportation) when assessing VR sickness. Chang et al. [6] presents a categorization model separating contributing factors into hardware, content, and user factors. But such factors are rarely investigated simultaneously with each other under a compound test scenario. Similarly, Kim et al. [20] and Haamer et al. [10] highlight that previous research has controlled variables to the point at which the results are no longer ecologically valid.

By simultaneously combining and experimentally manipulating a number of real-world variables mode of locomotion (web-swinging vs. jet packing), interaction complexity (shooting, throwing, grabbing), detail level of environment (high vs. low detail), and cognitive presence (with vs. without narrative context) under one experimental design, this study addresses this gap. This allows empirical investigation of the joint impact of these variables on discomfort and immersion in VR.

Alternative locomotion styles like arm-swinging provide a naturalistic physical engagement that may improve spatial awareness while reducing simulator sickness in certain users [7].

Homogeneous Participant Sampling

Existing research is likely to sample from a restricted population, often university students or individuals already familiar with VR equipment. Kim et al. [20] and Webb et al. [28] both

point out that different people are more or less susceptible to motion sickness depending on their age, gender, and prior experience with immersive equipment. Results will overestimate tolerance in the general user group and underestimate motion sickness susceptibility among novice or elderly users if diversity is not representative.

This research enhances generalizability by enrolling a demographically diverse participant sample (n=33) across gender, age, and levels of VR experience. This allows both increased appreciation of response variation among users and more generalizable VR design recommendations.

Prototype Simplicity and Unrealistic Interaction Scope

The majority of the VR sickness studies have small, unrepresentative testbeds of real game dynamics. For example, Haamer et al. [10] and Casas et al. [5] note that most current prototypes employ bounded interaction (e.g., point-and-click teleportation or fixed viewing) and lack a range of mechanics such as free-form flying, swinging, or gunplay.

To counter this, two tie prototypes are designed:

1. **Prototype 1** has low-fidelity graphics and actions like trash collection through web-swinging and jet packing in a low-fidelity sci-fi setting.
2. **Prototype 2** includes high-end features like air combat, dynamic enemies, realistic lighting, high-fidelity textures, and compelling storytelling.

These prototypes support side-by-side comparison of low and high interaction intensities and in what combinations most immersion is reached with the least sickness.

Underuse of VR-Specific Evaluation Metrics

Although the Simulator Sickness Questionnaire (SSQ) remains predominant, it is not responsive to modern VR interaction and HMD-based travel. Kim et al. [20] introduced the Virtual

Reality Sickness Questionnaire (VRSQ) as a better measure, highlighting oculomotor and disorientation symptoms, which are more pronounced in VR than in traditional simulators.

VRSQ and MSQ (Motion Sickness Questionnaire) are utilized in this experiment to quantify participants' discomfort during as well as after every game play session. This two-measure strategy offers a broader perspective of symptom generation in response to varied navigation and interaction demands.

Physiological data (e.g., pupil dilation, accuracy, HRV) also offer real-time insight into VR-induced discomfort and should complement questionnaires [22].

Static Game play Complexity in Previous Work

Real-world VR applications have higher task challenge and compound difficulty. University prototypes are static with pre-determined task difficulty that is oblivious to cognitive load throughout the duration. Chang et al. [6] and Haamer et al. [10] propose designs for increasing or adaptive difficulty levels that better simulate commercial game play patterns.

This study uses a phase-based challenge design:

1. Phase 1 introduces debris collection through intuitive web-swinging.
2. Phase 2 introduces danger from the enemy with the need for jet pack evasions and gun defense multitasking.

Tutorials are interpolated between phases to reduce cognitive load and give players gradual adaptation to novel mechanics.

Narrative Context and Story-Driven Motivation Gap

While VR sickness is extensively researched with technical contexts, the psychological impact of immersion is less examined. Narrative-based situations were found by Lonner et al. [24] to successfully reduce subjective discomfort with long-term cognitive interest and goal-directed

attention. Similarly, Haamer et al. [10] report the significance of incentives to avoid sensory conflict.

Some studies also suggest that legibility of mission-related text under various environmental visual effects impacts cognitive comfort and situational awareness [14].

This gap is filled by **Prototype 2** with an Intergalactic Ranger player who has to disable danger and restore order in a space sector. Voiceovers that serve a story, mission prompts, and in-context goals are used to bring user attention to bear, imbuing game play with motivational value.

No Temporal Adaptation Measures

Most recent studies treat VR discomfort as a single measure and do not examine the change in symptoms during phases of interaction. Nonetheless, studies like Kim et al. [20] reveal that symptoms decrease with time as participants get accustomed to motion, visuals, and tasks.

Engagement and motion sickness are quantified in this experiment at transitional critical phases:

1. After web-swinging tutorial
2. After jet packing and enemy combat session
3. Final debrief phase

This phased analysis provides both concurrent and cumulative feedback about adaptation to VR interaction over extended spans of time.

2.0.6 Filling the Gaps: Research Contributions

Through the use of immersive graphics, narrative-driven game play, naturalistic locomotion methods, and phased tutorials with two consecutive prototypes, this research fills many

significant gaps in current VR motion sickness research. These contributions are:

1. Holistic, multi-faceted analysis that synthesizes hardware, interaction, and narrative variables.
2. Demographically representative sampling of users.
3. Difficulty adaptation as a component of phase-based play.
4. Use of current VR-specific metrics of measurement (VRSQ, MSQ).
5. Systematic on boarding and training components for reducing initial-session discomfort.
6. Measurement of symptom presentation by game play phase.

Together, these results shed light on the effects of real game-playing situations on VR-induced motion sickness and offer design guidance supported by verification to next-generation immersive experience creators.

2.0.7 Research Questions

As per the literature review and understanding of shortcomings of previous research, this study formulates five important research questions to better understand, assess, and reduce VR-induced motion sickness in immersive VR environments:

1. **At the beginning, What are the primary hardware, navigation, and interface processes that contribute to motion sickness in virtual reality, and how do they impact user comfort and immersion?**

There is more than one explanation for the prevalence of VR motion sickness. Instead, it combines a variety of travel strategies (such as web-swinging, teleportation, or jet

packing), sophisticated interfaces (like shooting, tossing, or multitasking), and hardware constraints (like latency and field of view). Haamer et al. [10] and Casas et al. [5] show that smooth, natural motion patterns (like jet packing) may be less uncomfortable than broken or angular motion patterns (like web-swinging). Chang et al. [6] emphasize latency and field of view as the main sensory mismatches. Neuro scientific studies reinforce the theory that motion sickness stems from mismatches in sensory integration within the brain’s vestibular and visual systems [17][31]. This mismatch is perceived differently across age groups, as older adults may experience more intense discomfort [16]. This inquiry attempts to investigate how these technical and design elements work together to affect overall user comfort and immersion.

2. How do individual factors such as age, gender, and prior VR experience influence the susceptibility to motion sickness when engaging with a virtual environment?

User traits play a decisive role in determining VR tolerance. Kim et al. [20] demonstrate that females are more sensitive, whereas Webb et al. [28] demonstrate that novices and younger subjects are more uncomfortable. Additionally, Weech et al. [30] highlight that psychological and physiological traits such as balance ability, visual dependency, and anxiety levels may amplify cybersickness susceptibility beyond demographic categorization. This study brings demographic diversity to investigate how these traits interact with advanced navigation and interaction styles such as jet packing and object manipulation.

3. What are the effects of rapidly compounding game play difficulty on adaptation and symptoms of motion sickness during extended use of VR game play? as proposed by Haamer et al. [10] and Chang et al. [6], Gradually increasing the exposure of players to new challenges within game play allows adaptation and

minimizes disorientation. Step-by-step game play employed in this research, beginning with simple tasks and incrementally closing in on combat models of multitasking, is employed to determine if growing complexity aids adaptation physiologically and mentally between sessions.

4. **How do in-house-developed training modules and story-driven game play, with two aspects of immersive design, reduce discomfort while enhancing the VR experience?** Immersion has been shown to reverse the perception of discomfort via the prioritization of user attention on essential tasks. Lonner et al. [24] set out that framing narrative optimizes cognitive presence, and Haamer et al. [10] documented evidence that mission prompts and narrative reduce physiological symptoms. The experiment incorporates these aspects into IGR New Game play and tests their role in increased participation and comfort.
5. **How can design principles be developed to allow developers to create VR prototypes that maximize immersion and engagement with minimal motion sickness?** The final goal of this research is to suggest an empirically tested framework to VR developers. Implications will be made based on experimentation with VRSQ and MSQ tools [20], aligning best practices in navigation, visual design, training on boarding, and environmental immersion. Based on work conducted by Webb et al. [28] and Casas et al. [5], implications attempt to make their way to real-world deployments in education, training, and games.

Chapter 3

Methodology

The methodology used in designing, developing, and testing the virtual reality prototype, which is titled **IGR-Intergalactic Ranger**, is described in this section. The objective is to explore the role of locomotion mechanics, interaction complexity, and immersive design in VR-induced motion sickness according to previous research on VR discomfort and user experience.

3.0.1 Prototype Description

The IGR-Intergalactic Ranger prototype was constructed using Unity’s Universal Render Pipeline (URP) for high-visual-performance performance on standalone VR hardware. The game is played out in a zero-gravity science fiction outer space environment, with the player playing as a galactic ranger.

Two prototypes were created to test different design intensities:

Prototype 1: A minimalist version with only the locomotion mechanics of web-swinging and jet packing. The version has no training modules, narrative, or world description so that the study can control for physiological response to basic navigation systems.

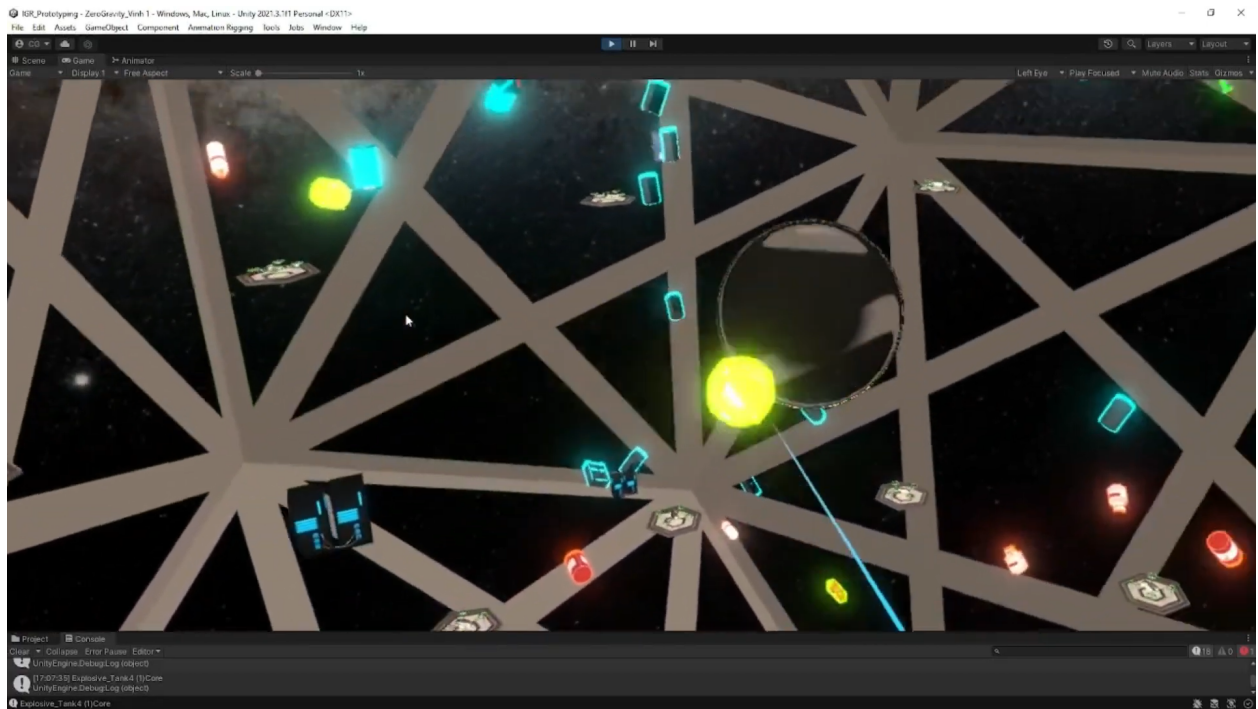


Figure 3.1: IGR Old Prototype with Grabbing Objects mechanism

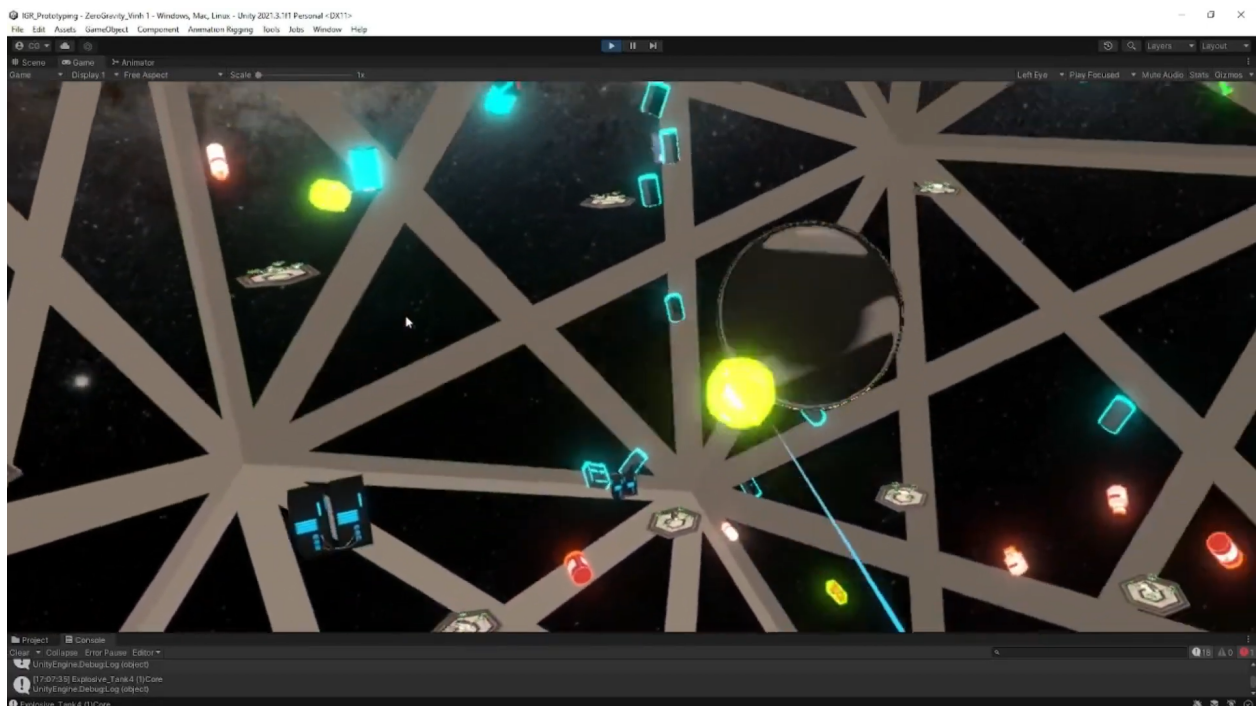


Figure 3.2: IGR Old Prototype with Grabbing Objects mechanism

Prototype 2: A more advanced and interactive version that incorporates realistic textures, ambient space elements, enemy attacks, mission objectives, and formal training modules. It is utilized to test the impact of immersive storytelling and task advancement on motion sickness and engagement across time. Figures 3.1 and 3.2 demonstrate the object-grabbing and web-swinging mechanism implemented in the prototype.



Figure 3.3: IGR New game play with Grabbing Objects mechanism

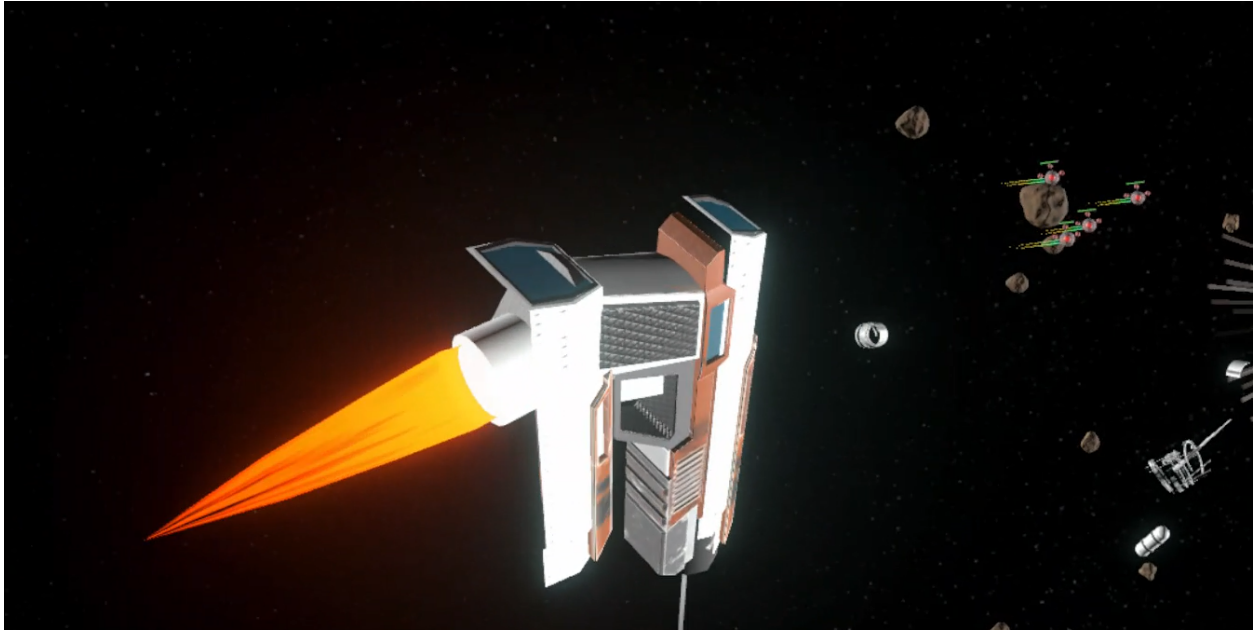


Figure 3.4: IGR New game play with jet packing mechanism



Figure 3.5: IGR New game play with Web-Swinging Mechanism



Figure 3.6: IGR New game play with Gun Shooting Mechanism

This contrast of prototype 1 and prototype 2 allows evaluation of sensorimotor mismatch across minimal and immersive environments, reflecting findings in [16], which emphasize age-related sensitivity and the need for adaptive environments to reduce motion mismatch. The approach is consistent with recommendations of Haamer et al. [10] and Webb et al. [28], both of whom underscore the importance of progressive, narrative context environments to minimize VR sickness and increase cognitive focus.

3.0.2 System Design

System design ensures tight interaction between user input, headset tracking, and environmental feedback to minimize latency and visual inconsistency factors proven to cause cybersickness [6][18]. To further mitigate discomfort, passive vestibular strategies such as stable orientation cues and smooth acceleration transitions were incorporated, informed by findings in [27]. Additionally, interaction design minimizes cognitive overload, echoing the

design recommendations from [22].

Hardware: The Meta Quest 3 stand-alone headset is utilized due to its high-resolution displays, low-latency motion tracking, and onboard processing. Such technical specifications are needed to ensure immersion while minimizing usual causes of motion sickness, such as motion-to-photon lag and FoV inconsistencies [5][28].

Input and Interaction: Hand and head tracking are integrated in a natural way for real-time interaction. Grip-based controls are used to naturally activate core mechanics like swinging, thrust-based jet packing, and object manipulation, which mimics natural motion as proposed by Kim et al. [20].

Real-time switching between game phases, monitoring user feedback, and collecting sickness data via both VRSQ and MSQ instruments at pre-programmed checkpoints are supported by the system.

Game Mechanics

The "Intergalactic Ranger (IGR)" prototype is developed for Meta Quest 3 and Quest 3s, using a wide range of immersive VR mechanics to test their impact on motion sickness and user accommodation. In the wake of recent research on naturalistic interaction and movement-induced discomfort in VR [5][10], the prototype will investigate the effect of integrated sensory input and challenge complexity on player experience. The following basic game mechanics are implemented:

1. **jet packing:** Players travel in a zero-gravity environment with unrestricted 3D movement provided by jet packs. The system is adapted for comfortable, physics-controlled movement, with users able to control speed and direction via the Quest controllers. Based on models of motion comfort in [5], jet packing avoids disorientation through sustained, non-jerking acceleration. To support motion predictability and reducevection-

related disorientation, velocity was incrementally scaled and consistent acceleration applied, as advised in [13].

2. **Web-Swinging:** Through using traversal mechanics that are fast-traversal-based, players can shoot webs onto space architecture and web-swing through vast distances. This kinetic system achieves high momentum with difficult motion sickness thresholds while being very smooth. Casas et al. [5] categorize this as a high-intensity traversal strategy requiring supplementary proper orientation information and environmental anchors to reduce visual-vestibular conflict. Environmental reference points were emphasized to anchor user perception and reduce sensory mismatch in fast traversal, aligned with motion-sickness mitigation techniques highlighted in [8].
3. **Gun Shooting:** Objects or enemies are fired at by players with weapons that are triggered through VR controllers. The shooting mechanism is interactive so that immersion and player agency in combat are not compromised, as suggested by Haamer et al. [10] as a key factor of immersion and goal-directed presence.
4. **Trash Grabbing and Throwing:** It is a mechanism that entails stretching out, grabbing trash, and throwing it into target reactor cores. It can be used for fine-motor interaction testing, proprioception, and physicality of VR tasks. This mechanism aligns with Kim et al. [20], who emphasize oculomotor strain in repetitive and precision object manipulation.

Prototype 1: features trash grabbing and throwing, jet packing, and web-swinging in a low-fidelity setting with no visual effects or high-level interaction. The user gets to perform a time-constrained task of picking up and throwing space debris into the core. This prototype is a foundation for motion sickness with low interactivity.

Prototype 2: in contrast, integrates jet packing, web-swinging, firing, and multi-modal interaction in an abundant environment. The user gets to perform a time-constrained task

of picking up and throwing space debris into the core, but with the complexity that grows over time, adhering to adaptive game play models proposed in [6].

Combat Mechanics

Fight dynamics of IGR New game play involve shooting and target shooting with weapons using Quest 3/3s controllers. Dynamic movements of enemies and evading them using jet packs or eliminating them using target shooting are things that involve decision-making under movement, as opposed to linear target shooters, whose behavior adjusts to require strategic decision-making under movement. This is informed by Lonner et al. [24], who showed the way dynamic task goals reduce subjective sickness and increase cognitive attention.

Battle build-up becomes more intense cumulatively to test physiological response under increasing stress, as Haamer et al. [10] suggest. Gun recoil and aiming were tuned to avoid abrupt head motions, in line with vestibular calibration strategies from [27], enhancing comfort during reactive combat.

Increasing Difficulty and Training Modules

Two controlled, video-supported training modules are incorporated to help participants gradually become accustomed to the VR experience and reduce initial disorientation:

Tutorial 1 introduces players to *web-swinging*, *trash grabbing*, and *throwing*. It is located in a serene, zero-distraction environment where mechanics are demonstrated via video tutorials and then practiced interactively.

Tutorial 2 focuses on *jet packing*, *gun shooting*, and redefining the trash-handling tasks in more pressurized environments. The module prioritizes time-constrained task completion, and users are encouraged to learn mechanics while managing their level of comfort.

The gradual layering of complexity over sessions aims to prevent early-stage overload,

consistent with cognitive adaptation principles reported by [22]. Such two-stage onboarding is consistent with best practices documented in [10][28], where mechanical exposure for an extended period enables sensorimotor tolerance and skill development in tasks.

Storytelling and Environmental Immersion

To create real-world applications where user interaction is paramount (e.g., games, education, simulation), the game combines environmental storytelling and visual richness. The science fiction environment includes highly stylized sci-fi visuals, dynamic lighting, and interactive objects that react to the player’s actions.

InIGR New game play, the player becomes an ”Intergalactic Ranger” who wants to retake control of an invaded space station. Voiceover dialogue, mission prompts, and ambient audio are used to generate a narrative loop that sustains cognitive interest.

As shown in Lonner et al. [24] and supported by the work of Haamer et al. [10], narrative immersion keeps players focused on goals and reduces their susceptibility to discomfort via attention anchoring on high-level objectives.

All players are educated with clear onboarding protocols and safety instructions and made aware that their responses will be anonymized. They were also made aware of game play protocols, length of sessions, and headset adjustment and provided access to tutorial videos before game play.

his design encourages attentional engagement, a mechanism shown to reduce sickness by cognitively anchoring user focus away from sensory mismatch [29].

3.0.3 Data Collection

Assessing motion sickness through VR gaming involves a mixed-methods approach, which combines quantitative questionnaires with physiological monitoring and qualitative results. All players go through three VR sessions:

1. **Prototype 1:** A world with low complexity and easy object grasping and trash-throwing timed challenges without visual feedback or advanced traversal mechanics.
2. **Prototype 2:** A high-fidelity world with advanced features such as jet packing, web-swinging, shooting from guns, and trash pickup/throwing in interactive combat scenarios.
3. **Swarm VR:**

Swarm VR: Benchmark Application and Comparative Role

Swarm VR is a commercially available VR game with active, high-speed movement and action-centric game play. It has a very active movement system in which users swing through environments using arm-based movement, similar to Spider-Man mechanics. The player freely navigates three-dimensional space, usually undergoing high-speed battles with enemies and constant changes in direction. These are defined contributing elements to vestibular load, which is a primary element in producing virtual reality sickness [6][28].

For the sake of this research, Swarm VR served as a field test standard to judge motion sickness impact against two in-house-built versions: IGR Old and IGR New. The original IGR prototype (IGR Old) was straightforward, with the focus primarily being on basic locomotion and object interaction without any extensive graphical or storyline features. It included basic mechanics such as picking up and tossing objects, with little visual stimuli and no scripted tasks. Although the simplicity made it suitable for baseline analysis, it lacked immersion and challenge-based elements, which minimized stimulation of motion sickness relative to more active commercial simulators [20][18].

Swarm VR provided a practical counterexample by virtue of its entrenched position in

the gaming community and its established interface that inherently increases the user’s sensory input along the visual, vestibular, and proprioceptive axes [25][24]. Comparison against IGR Old was intended to show the flaws of basic locomotion in inducing discomfort reactions and emphasize the need for more immersive environmental and game play factors when measuring true cybersickness risk.

On the other hand, the **IGR New** prototype was built with the specific intention of introducing these exact observations. It introduced advanced locomotion via jet packing and web-swinging, task-based narrative, richer worlds, and richer audio to improve spatial presence. These changes made IGR New closer to Swarm VR with regards to pacing and sensorial complexity. Therefore, it was a better analogue to test the implications of factors inducing motion sickness in modern VR environments [5][17][31].

By incorporating Swarm VR into the experiment pipeline, the present research ensured that the testing was not limited to in-house applications but also tested against real-world design standards, making the findings more valid and reliable [12][15].

Training modules are offered for both prototypes. Tutorial 1 touches on web-swinging, trash grabbing, and timed throwing exercises. Tutorial 2 introduces jet packing, gun firing, and complex trash handling. These modules are made available through in-game instructional videos to facilitate uniform learning.

Quantitative Data Collection:

3.0.4 Motion Sickness and VR Questionnaires in Assessment

In order to effectively assess the motion sickness tendency of players and VR-induced symptoms during the game study, two validated questionnaires were employed: the Virtual Reality Sickness Questionnaire (VRSQ) and the Motion Sickness Questionnaire (MSQ).

MSQ is employed as a baseline diagnostic tool to quantify an individual’s general pre-disposition to motion sickness. It tallies past symptoms occurrence across typical modes of daily transportation like traveling in a car, airplane, or boat. MSQ’s utility lies in giving a participant’s previous inherent sensitivity level prior to exposing them to VR stimuli so that researchers can tease apart inherent tendencies from voluntary discomfort responses specific to VR [25][31].

In contrast, the VRSQ is a more recent and VR-specific questionnaire that measures physiological and sensory symptoms experienced during or after VR interaction. The VRSQ is designed to be concise yet sensitive to salient VR-related symptoms and includes items measuring oculomotor and disorientation symptoms such as dizziness, blurry vision, and eye strain without requiring full simulator configurations like the older SSQ [20]. Its design makes it particularly ideal for rapid, in-field assessments in both research and commercial settings.

Comparative research has also attested to the high reliability and specificity of VRSQ, with correspondence to more established measures such as the Simulator Sickness Questionnaire (SSQ), but with greater practicality for use in VR studies [2]. Furthermore, newer alternatives such as the Cybersickness in Virtual Reality Questionnaire (CSQ-VR) have been contrasted with and validated against the VRSQ, with high correspondence and indicating the continued utility of the latter as a fundamental measurement instrument [12][21].

In recent research, such subjective measures are routinely cross-validated against cognitive and physiological measures i.e., reaction time, performance variability, and user self-report under stress to assess more general immersion-induced workload. For instance, physiological stress responses and cognitive load variability in the course of immersive simulations have been correlated with the onset of motion sickness, highlighting the utility of instruments like VRSQ for early detection and iterative design [11][22].

Augmenting standardized questionnaires, observational data is key to in-context measurement of user discomfort. Adjustment of the headset, abnormal pausing during play,

or evasion of particular locomotion techniques are all indicative signs of hidden discomfort. Studies have found that such behavioral signals are often in line with self-reported symptoms and can serve as early indicators of rising cybersickness or disengagement [26][9].

Together, these methods provided a mixed-methods, holistic foundation for determining the effectiveness and comfort level of the proposed VR prototypes.

Open-ended Feedback Collection: Participants provide free-text comments at each phase, citing discomforts, communication ease, and preference for various prototypes. Field observation is also recorded with attention to behavior patterns such as incessant headset adjustment, stalling, or task deferment, which can be linked to the levels of discomfort [6].

Evaluation Metrics

A two-layered evaluation framework is used to investigate user experience:

Subjective Measures:

1. **VRSQ** [20] is the main real-time measurement tool, capturing discomfort during game-playing periods. It targets oculomotor symptoms, as opposed to nausea-filled measures, and is best used for VR headset configurations.
2. **MSQ** [18] is the accompanying measurement that identifies predisposed subjects, enabling comparative analysis before and after game playing.

3.0.5 Measures of Evaluation

To effectively measure the incidence and severity of VR-induced motion sickness across different stages of game play, this study employs a multi-dimensional evaluation paradigm. This includes subjective questionnaires and objective quantifications to capture the full spectrum of user discomfort, adaptation, and immersion. The objective is not just to identify

symptoms but to chart how they change with game play mechanics, training exposure, and interpersonal variability.

Subjective Measures

1. Virtual Reality Sickness Questionnaire (VRSQ)

The VRSQ by Kim et al. [20] targets symptoms specific to VR such as oculomotor fatigue and disorientation, specifically. Unlike the older SSQ, it does not contain nausea; thus, it is more appropriate to use while comparing sickness in HMD-based systems. The subjects complete the VRSQ after each main game play session (Swarm VR, IGR Old Prototype, and IGR New game play), facilitating phase-specific symptom tracking between different interaction types such as jet packing, swinging, and shooting.

2. Motion Sickness Questionnaire (MSQ)

The MSQ, constructed by Kennedy et al. [18], is administered at the start of the study to assess a participant's overall susceptibility to motion sickness. The baseline is utilized to place responses to the VR experience into context, specifically identifying participants with greater susceptibility to discomfort in immersive or engaged game play segments. The same application has been warranted in studies such as Webb et al. [28].

3. Verbal Reports and Open-Ended Feedback

In addition to official questionnaires, participants also give their comments and reflections after each game play experience. These comments usually reveal undemonstrated discomforts, emotional reactions, or interruptions of immersion that would not be elaborated on in default formats. In line with the recommendation of Lonner et al. [24], qualitative feedback here is beneficial for grasping user experience in VR environments holistically.

Objective Experiments

3.0.6 Phased Evaluation Protocol

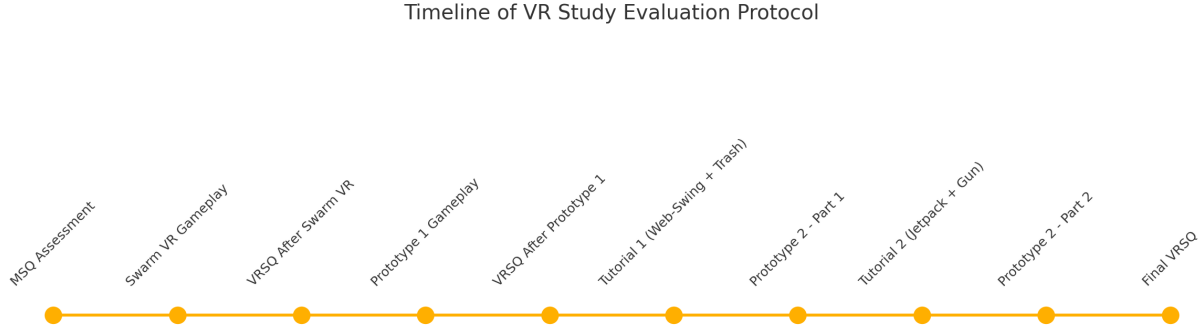


Figure 3.7: Timeline of phased VR game play and evaluation protocol. Each stage illustrates the flow from baseline assessment to game play phases, with sickness questionnaires administered after each session.

To ensure sickness and immersion levels are tested under true game play conditions, the study is divided into a structured multi-phase protocol. Each phase is performed with increasing interaction complexity, from passive benchmark to advanced, in-house-devised VR mechanics, along with strategic training sessions for adaptation measurement.

The protocol is carried out with 33 participants, who have been recruited based on extremely stringent requirements regarding age, gender, and VR experience. Informed consent forms were handed out to all participants beforehand, and the questionnaires are anonymized and handled in complete confidentiality.

1. **Initial MSQ Assessment** Before any gaming, participants complete the Motion Sick-

ness Questionnaire (MSQ) [18] to establish a baseline measure for susceptibility. It permits VR-specific distress and sensitivity to be distinguished.

2. **Swarm VR Benchmark game play** Participants undergo a brief trial of Swarm VR a commercial arcade shooter that is noted for its active movement and fast firing. These characteristics, while engaging, are also documented to intensify sensory mismatch and discomfort due to high movement realism [9]. It serves as a benchmark to compare the impact of commercial play against that of in-house prototypes. Swarm VR’s motion profile offers an effective testbed for vection sensitivity, especially under passive motion exposure [13].
3. **VRSQ After Swarm VR** Immediately after the Swarm VR session, the first VRSQ [20] is administered to identify initial symptoms of eye strain, dizziness, or disorientation.
4. **Prototype 1 game play** The participants then experience IGR Old Prototype, a bare-bones trash-throwing simulation without special effects, navigation controls, or enemies. It is intentionally minimalist to serve as a low-motion baseline for our study.
5. **VRSQ: Following IGR Old Prototype** VRSQ is given again for comparison of symptoms owing to this minimalist, task-oriented game play.
6. **Prototype 2 Training Tutorial 1 (Web-Swinging + Trash Interaction)** Participants then engage in an interactive video tutorial to get accustomed to web-swinging and trash interaction mechanics. The tutorial also sets clear expectations for task completion within a time constraint.
7. **Prototype 2 Part 1 game play** The players are introduced to IGR New game play, which has engaging visuals, enhanced mechanisms such as swinging, and adap-

tive difficulty. It provides a challenging experience with feedback loops to mimic the progression of the game.

8. **Prototype 2 Training Tutorial 2 (jet packing, Gun Shooting, Combact and Trash Interaction)** Here, the players are trained in handling the jet pack and gun and doing complex trash jobs under time limits. This is to prevent onboarding sickness and mental load when playing high-complexity games.
9. **Prototype 2 Part 2 game play** The game play continues with additional game mechanics such as gun firing, combat with enemies, and navigation. This stage is meant to mimic actual VR conditions of multitasking, exploration, and fighting.
10. **Final VRSQ Measurement** The final VRSQ is administered after all tutorials and game play modules have been completed to assess total motion sickness symptoms and adaptation after prolonged exposure.

This staged design allows us to examine not just when and under what conditions VR sickness occurs, but how subjects learn as complexity accumulates stepwise. Including tutorials ensures perceived discomfort is not due to unfamiliarity with the controls but due to true motion sensitivity. On top of that, tutorial sequences were carefully timed to enable progressive adaptation, in line with the step-wise immersion recommendations in [22]. The method is in line with prevailing best practice in VR testing, as emphasized in recent reviews by Chang et al. [6], Haamer et al. [10], and Casas et al. [5].

Assessing motion sickness through VR gaming involves a mixed-methods approach that includes quantitative, qualitative, and optional physiological metrics. Participants undergo three VR sessions: IGR Prototype 1, Swarm VR, and IGR Prototype 2.

- **MSQ (Motion Sickness Questionnaire)** Administered before game play to assess

baseline susceptibility.

- **VRSQ (VR Sickness Questionnaire)** Administered after each game play phase to measure symptoms.

Both instruments were deployed using **Qualtrics Experience Management** and collected anonymously. The datasets were exported in CSV format and merged by **Participant_ID** in SPSS v29 (wide format: one row per participant, multiple columns for different conditions). All physiological and performance data align with recommendations for cybersickness and cognitive load tracking as described in [22].

MSQ Variable Mapping

Table 3.1: MSQ Question-to-Variable Mapping

Survey Question	Variable Name
How often do you use VR headsets?	VR_Use_Freq
Have you ever experienced motion sickness (car, boat, airplane)?	MotionSick_History
Have you previously experienced motion sickness while using VR?	VRSick_History
What type of VR games have you played before?	VR_Game_Types
Rate your comfort level with fast-moving visuals in VR:	FastVisual_Comfort
Do you wear glasses or contacts while using VR?	Wear_Glasses_VR
Continued on next page	

Table 3.1 – continued from previous page

Survey Question	Variable Name
How easily do you adapt to new control schemes in VR?	Adapt_New_Controls
Are you familiar with the concept of jet packing or swinging in VR?	Know_Jet_Swing
How sensitive are you to rapid head movements in VR?	HeadMove_Sensitivity
Have you used Meta Quest 3 / 3S before?	Used_Quest3
Rate your vision clarity in VR (e.g., no blurriness):	Vision_Clarity_VR
Do you feel any eye strain after 10 minutes of VR usage?	EyeStrain_After10
Have you previously played games with free locomotion?	FreeMove_Experience
Do you feel confident identifying if you are experiencing VR sickness?	Identify_VRSickness_Conf
Do you have any history of migraines or vertigo?	Migraine_Vertigo_Hist

VRSQ Variable Mapping

Table 3.2: VRSQ Variables and Types

Measurement Focus	Variable Name
Movement intensity	Move_Intensity
Dizziness or disorientation	Dizzy_Disorient
Eye strain	Eye_Strain
Visual overwhelm	Visual_Overwhelm
Nausea or discomfort	Nausea_Discomfort
Confused navigation	Confused_Navigation
Would you continue in VR?	Comfort_VR_Continue
Comfort during movement	Move_Comfort
Main discomfort element	Discomfort_Element
Latency or lag noticed	Latency_Notice
Confidence while aiming	Aim_Confidence
Balance challenge during play	Balance_Challenge
Cause of motion discomfort	Move_Discomfort_Type
Level of immersion	Immersion_Level
Overall experience rating	Overall_Experience
game play condition	Condition

3.0.7 Measures of Evaluation

Subjective Measures:

- **MSQ** - Assesses baseline motion sickness sensitivity before game play.
- **VRSQ** - Administered after each session to detect symptoms during or post game play.

- **Open Feedback** - Collected after each phase to capture emotion, discomfort, or game play suggestions.

Objective Measures (Optional):

- **Motion Tracking** - Positional and rotational data from Meta Quest 3 logged for unusual movement patterns.
- **Physiological Monitoring (HRV/EDA)** - Optional wrist sensor data indicating stress responses to intense phases.

3.0.8 Analysis Methodologies

All data were analyzed in **SPSS v29** and visualized in **Excel 365**.

Descriptive Statistics

To summarize each conditions symptom reports:

- Analyze > Descriptive Statistics > Explore
- Factor = Condition, Variables = Immersion_Level, Comfort_VR_Continue, etc.

Boxplots and means were generated for each variable grouped by game.

Friedman Test

Used for Likert variables across 3 conditions:

- Analyze > Nonparametric Tests > Related Samples > Customize Tests
- Test: **Friedman**
- Variables: Immersion_Level, Comfort_VR_Continue, Move_Comfort, etc.

McNemars Test

Applied to Yes/No variables:

- Analyze > Descriptive Statistics > Crosstabs > Statistics > McNemar
- Used for: Eye_Strain, Dizzy_Disorient, Latency_Notice

Spearman Correlation (MSQ vs VRSQ)

Examined relationships between baseline motion sickness history and post-game discomfort:

- Analyze > Correlate > Bivariate > Spearman
- Variables: MotionSick_History, VRSick_History vs VRSQ metrics
- Merged datasets using Participant_ID

Thematic Analysis

Participants open-text responses were reviewed to identify recurring themes:

- jet packing or visual discomfort
- Difficulty in swinging
- Interface suggestions

Visualization Strategy

SPSS and Excel were used to create

- **Bar Graphs** - Mean comfort or immersion by condition
- **Heatmaps** - Correlation matrix of MSQ vs VRSQ

Each figure is referenced in the Results chapter for clarity.

Chapter 4

Experiment

4.0.1 Experimental Design

This research applies a phased experimental approach to determine the effect of various VR locomotion and interaction mechanics on motion sickness and immersion. Two prototype systems (Prototype 1 and IGR New game play) and one off-the-shelf VR game title (Swarm VR) are used to evaluate participant response across varying motion and interaction paradigms [1]. Mechanics range from basic object manipulation to high-speed travel and combat, allowing controlled comparison under escalating complexity [6][5]. The three-stage design captures motion- and interaction-induced sickness with respect to sensory mismatch, user adaptation, and cognitive engagement, as demonstrated in [16][22].

4.0.2 Experimental Setup

Hardware and Equipment

The experiment uses Meta Quest 3 and Quest 3s headsets. Standalone VR headsets with high-fidelity graphics, accurate hand/controller tracking, and motion telemetry in real-time offer freedom of movement and natural motion within a 6DoF environment ideal for assessing

mechanics like jet packing and web-swinging. Meta Quest 3s built-in motion prediction and hand tracking allow for reliable head movement capture, aligning with requirements outlined by [27] for vestibular feedback studies.

Software Environment

The interactive play environment, or *IGR - Intergalactic Ranger*, is deployed using Unity URP. IGR Old Prototype is characterized by simple trash-grabbing and throwing behavior with minimal effects. IGR New game play has advanced mechanics (jet packing, shooting, and web-slinging) and growing difficulty levels. Video tutorials within the application assist each level. Gradual mechanic integration was designed to prevent overload and mirror natural learning curves, following findings from [22] on task difficulty pacing in immersive environments.

Testing Space and Environment

All the testing is done inside a soundproof, ventilated room within the Computer Science building at Cal Poly Pomona. COVID-19 precautions include sanitizing VR hardware, making masks available, having hand sanitizer stations, and staggered scheduling to avoid overcrowding. The setup leaves enough room for free movement without being untethered but places emphasis on the safety of participants.

4.0.3 Procedure

Pre-Experiment Setup

Participants receive an email with a formal invitation email with a PDF consent form that details the purpose of the study, risks, and rights of participants. The consent form is read and signed under the university’s Institutional Review Board (IRB) process. Surveys

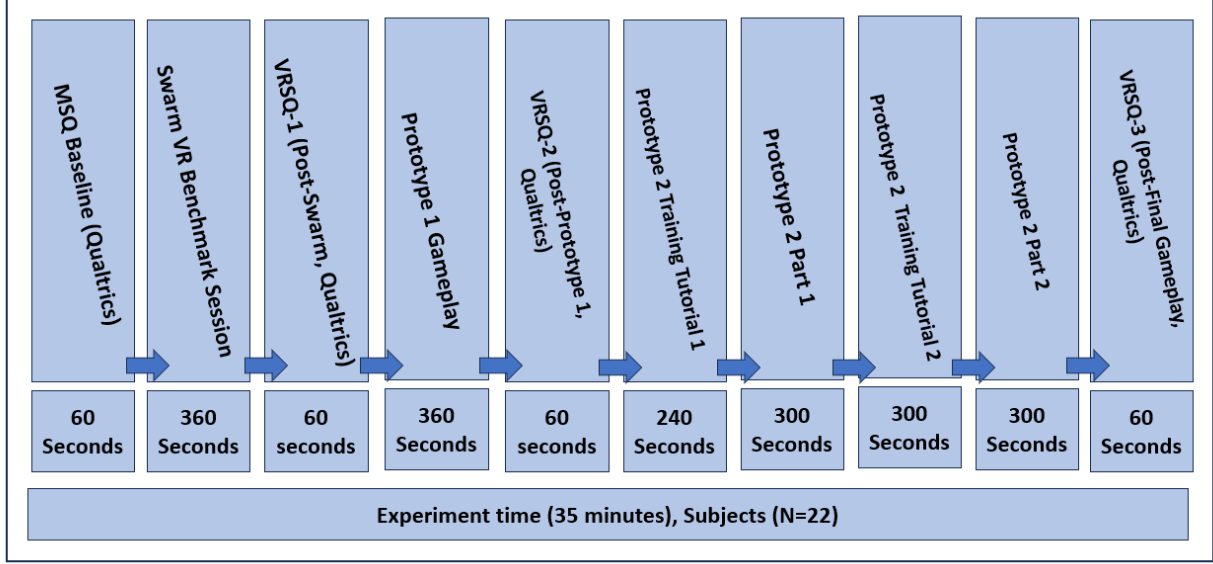


Figure 4.1: Experiment setup and timeline

like demographic forms, MSQ, and gathering feedback are stored securely on Qualtrics. Participants are informed that all information will be anonymized, stored securely, and used solely for academic research.

Participation is completely voluntary. Participants are free to discontinue or withdraw from study at any time without penalty. Participants who successfully complete the experiment are rewarded with a \$25 Amazon gift card for time and effort invested.

game play Phases

Each participant experiences the following experimental procedure (approximately 35-40 minutes): The experiment workflow’s visual demographic is provided by 4.1. Participants are given instructions on how to play the VR experience, and the experiment is well-structured and timed.

1. **MSQ Baseline (Qualtrics)(1 min)** The Motion Sickness Questionnaire (MSQ) [18] is administered to the participants at first to assess baseline motion sickness suscepti-

bility.

2. **Swarm VR Benchmark Session (6 min)** The participants are required to play Swarm VR, a commercial game used as a benchmark for the level of motion and the level of sickness. Swarm VR contains fast-paced, externally-controlled motion ideal for testing vection sensitivity and user passive adaptation, as described by [13].
3. **VRSQ-1 (Post-Swarm, Qualtrics) (1 min)** The Virtual Reality Sickness Questionnaire (VRSQ) [20] is administered to measure initial levels of discomfort.
4. **Prototype 1 game play (6 min)** The first prototype contains trash-throwing activity only in a low-immersion environment with minimal visual effects.
5. **VRSQ-2 (Post-Prototype 1, Qualtrics)(1 min)** Administered immediately after game play to track symptom onset.
6. **Training Tutorial 1 (5 min)** Provides web-swinging and trash interaction via video tutorial. Trainees undergo timing-based task completion.
7. **Prototype 2 Part 1 (4 min)** Players experience rich worlds through web traversing, combat, and adaptive trash game play.
8. **Training Tutorial 2 (5 min)** Provides jet packing and gun-shooting mechanics via video tutorial.
9. **Prototype 2 Part 2 (5 min)** Encompasses full traversing, shooting, enemy fighting, terrain challenges, and increasing difficulty.
10. **VRSQ-3 (Post-Final game play, Qualtrics) (1 min)** End-stage symptom gathering after full engagement with immersive mechanics.

Participant Guidance and Monitoring

Facilitators guide participants through each step using screencasting from headsets to monitor, ensuring that they are informed of instructions, safety rules, and their right to withdraw. Participants are observed for distress and are encouraged to withdraw if needed. There is a cooldown space with chairs and water. This design reflects a holistic understanding of sickness factors-from vestibular desynchronization to attentional anchoring-highlighted in [27][29].

Sample Size and Justification: The study has 33 participants, a sample sufficient to measure within-subject variation across five game play periods, consistent with studies by Kim et al. [20] and Casas et al. [5]. The sample is heterogeneous for gender, VR experience, and age (18-30) and generalizable to a variety of user populations in VR interaction studies.

Chapter 5

Results and Analysis

5.0.1 Data Analysis

5.0.2 Descriptive Statistics

Data analysis was conducted with IBM SPSS Statistics Version 29.0. Participant responses were collected through Likert-based and binary measures for primary motion and interaction variables for the three game play conditions: Swarm VR (benchmark), Prototype 1 (IGR Old), and Prototype 2 (IGR New game play). Qualtrics wide-format exported data were reshaped to long format within SPSS for repeated-measures analysis.

SPSS's "Analyze > Descriptive Statistics > Descriptives" and "Explore" features were utilized to calculate central tendency (mean), dispersion (standard deviation), and range (minimummaximum) between conditions for the Virtual Reality Sickness Questionnaire (VRSQ) and other variables like immersion, navigation comfort, and confidence indicators.

Descriptive statistics were first summed across game play sessions to derive overall trends in user experience and motion sickness susceptibility.

Table 5.1: Summary of Descriptive Statistics for Key Variables (Combined All game play Sessions)

Variable	Mean	Std. Dev	Min	Max
Immersion Level	3.94	0.76	1	5
Move Intensity	3.62	0.91	1	5
Balance Challenge	2.85	1.03	1	5
Dizzy/Disorient (Yes=1)	0.49	0.50	0	1
Confused Navigation (Yes=1)	0.37	0.48	0	1
Comfort: Would Continue	3.88	0.82	1	5

These significant findings suggest that users rated the experience as immersive ($M = 3.94$) and were open to playing VR more ($M = 3.88$) despite moderate self-reports of motion intensity ($M = 3.62$) and balance issues ($M = 2.85$).

Table 5.2: Descriptive Statistics by game play Condition

Variable	Condition	Mean	Std. Dev	MinMax
Move Intensity	Swarm VR	3.58	0.97	15
	IGR Old	2.79	0.93	14
	IGR New	3.00	1.06	15
Move Comfort	Swarm VR	3.70	1.05	15
	IGR Old	4.12	0.65	25
	IGR New	3.52	0.83	25
Aim Confidence	Swarm VR	3.21	0.74	24
	IGR Old	3.03	0.73	14
	IGR New	2.64	0.82	14
Balance Challenge	Swarm VR	3.18	1.07	25
	IGR Old	3.79	0.99	25
	IGR New	3.76	0.93	25
Immersion Level	Swarm VR	4.33	0.74	35
	IGR Old	3.91	0.58	35
	IGR New	3.88	0.82	25

Table 5.3: Interpretation of Mean Values by Performance Range

Mean Score Range	Interpretation
1.00 1.99	Very Low / Minimal
2.00 2.99	Low to Moderate
3.00 3.49	Moderate
3.50 3.99	ModerateHigh
4.00 4.49	High
4.50 5.00	Very High

Overview and Interpretation of Results

SPSS outcomes reveal a number of insights regarding patterns of game play-related comfort and discomfort when playing under conditions:

1. **Swarm VR** exhibited the highest overall immersion ($M = 4.33$) but also the highest movement intensity ($M = 3.58$) and balance challenge ($M = 3.18$). These are indicative that while the experience was immersive, it perhaps posed a higher risk of cybersickness, especially during rapid enemy combat maneuvers or environmental motion.
2. **IGR Old (Prototype 1)** offered most movement comfort ($M = 4.12$) and minimum movement intensity ($M = 2.79$), i.e., a stationary experience with reduced sensory stimulation. It also gauged lesser aiming challenge and disorientation but slightly lower immersion ($M = 3.91$) due to its primitive mechanics.
3. **IGR New (Prototype 2)** showed a well-balanced mix: medium movement intensity ($M = 3.00$), comfort ($M = 3.52$), and competitive immersion ($M = 3.88$). The rise in balance challenge ($M = 3.76$) is to be expected given the introduction of jet packing and web-swinging mechanics, which contribute to complexvection and camera movement.

4. **Aim Confidence** was lowest in IGR New ($M = 2.64$), possibly because of new control configurations, followed by IGR Old ($M = 3.03$) and Swarm VR ($M = 3.21$), which may have benefited from improving market-tested interfaces.

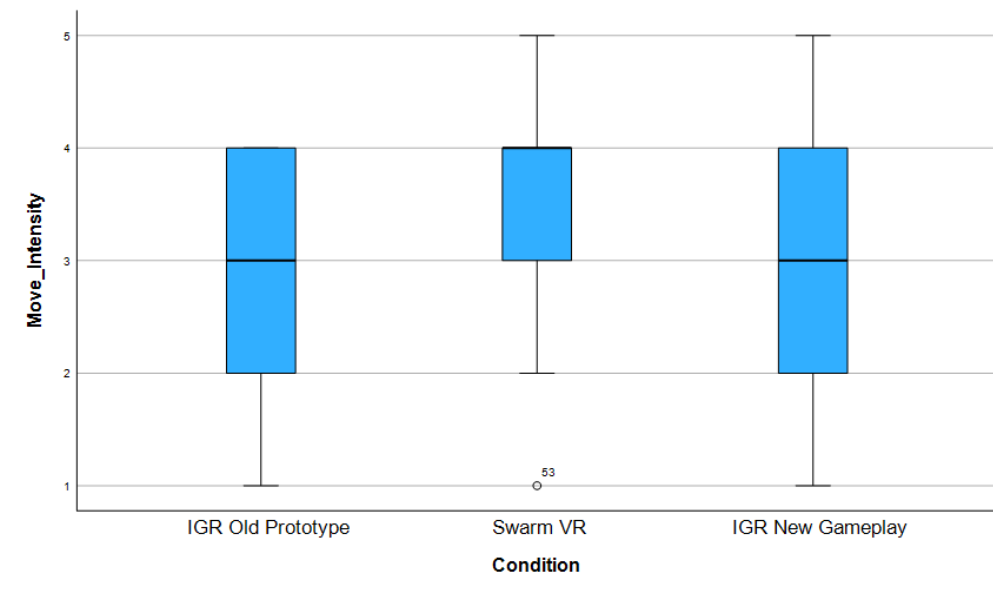


Figure 5.1: Box plot showing the distribution of **movement intensity** across game play conditions. The X-axis represents each VR condition (Swarm VR, IGR Old, IGR New), while the Y-axis represents perceived movement intensity rated on a 5-point Likert scale (1 = Very Low, 5 = Very High). Swarm VR has the highest median and widest spread, indicating higher and more variable movement; IGR Old is more grounded with minimal variation, while IGR New displays moderate intensity with wider variation due to mixed locomotion.

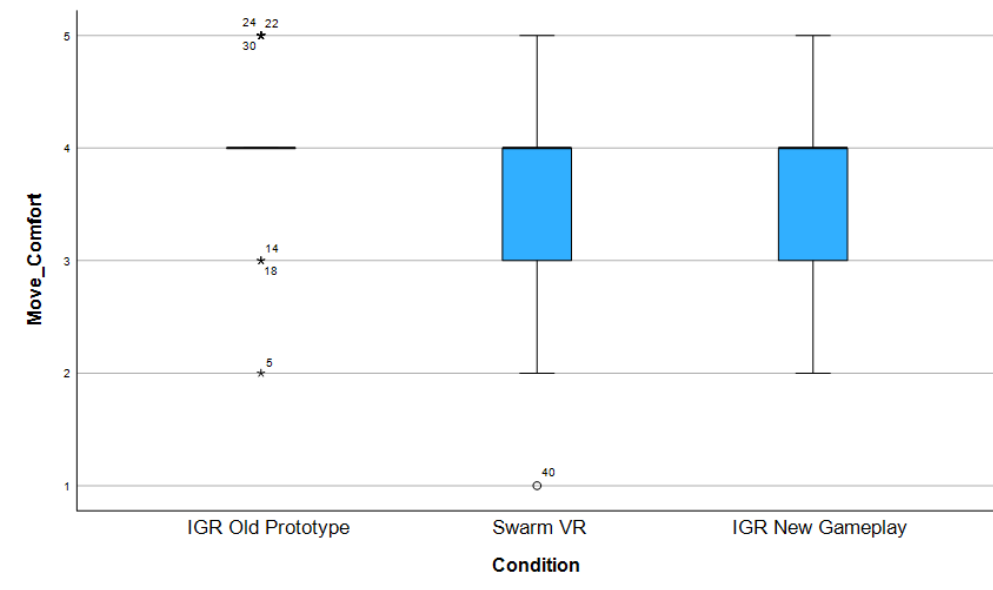


Figure 5.2: Box plot comparing **movement comfort** across the three VR experiences. X-axis shows game play conditions; Y-axis shows comfort level (1 = Not Comfortable, 5 = Very Comfortable). IGR Old has the highest median comfort with low variance, indicating more consistent and pleasant movement mechanics. Swarm VR shows moderate comfort with slightly higher variability, while IGR New shows good comfort but with greater spread, possibly due to fast-paced jet pack or web-swing dynamics.

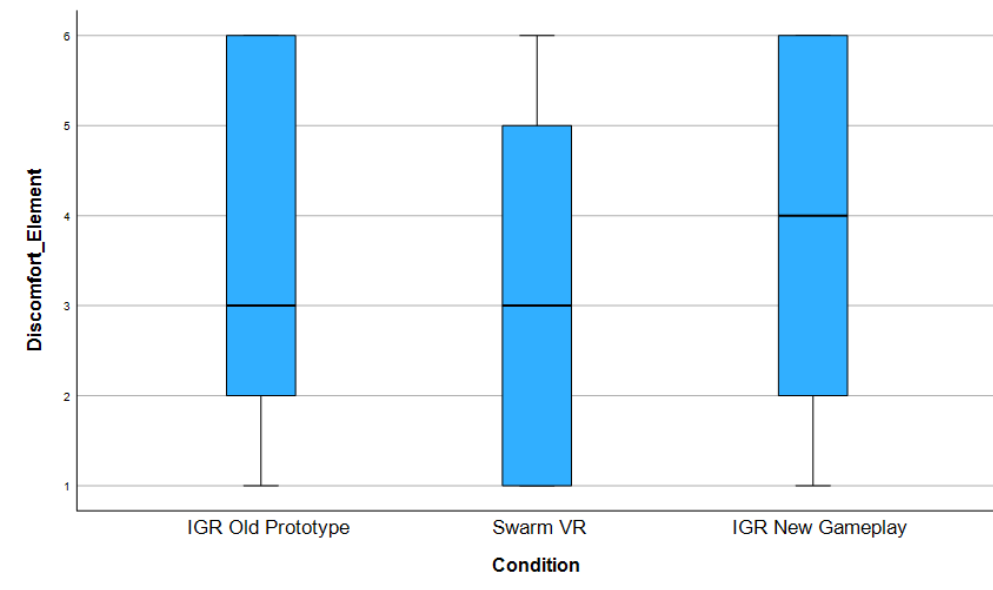


Figure 5.3: Box plot visualizing the **primary source of discomfort** reported across game play types. The X-axis lists the VR conditions and Y-axis shows discomfort level (1 = No Discomfort, 5 = High Discomfort). IGR New reveals a slightly higher median discomfort score, possibly linked to vection and navigation challenges. Swarm VR and IGR Old show tighter distributions around moderate discomfort levels.

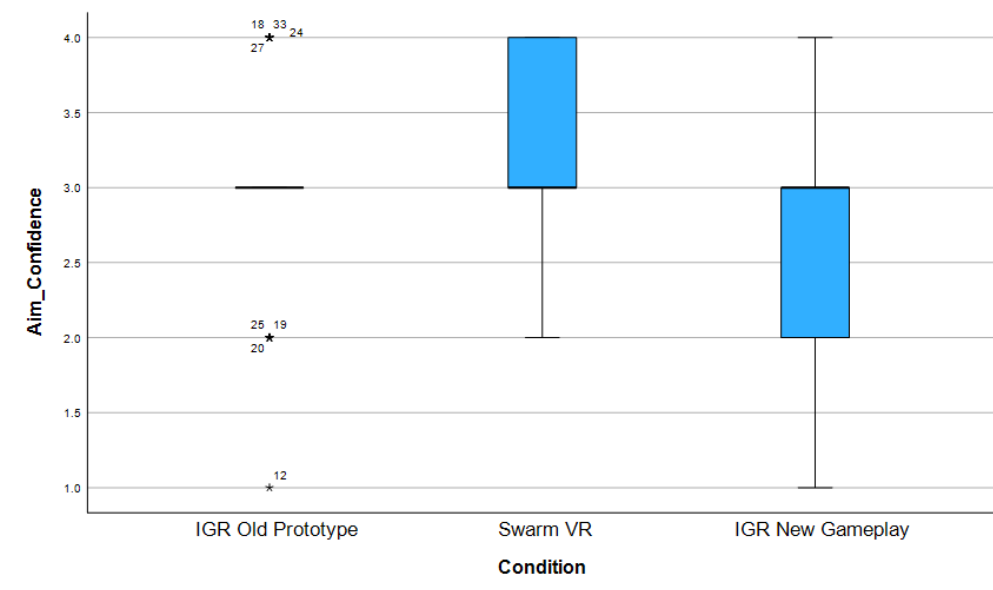


Figure 5.4: Box plot showing **aim confidence** levels across game play types. The X-axis reflects game play conditions, while the Y-axis denotes player confidence in aiming tasks (1 = Not Confident, 5 = Highly Confident). Swarm VR has the highest median aim confidence with narrow spread, reflecting polished mechanics. IGR Old shows average confidence with some variability, whereas IGR New has the lowest aim confidence, potentially due to novel mechanics like floating and multitasking.

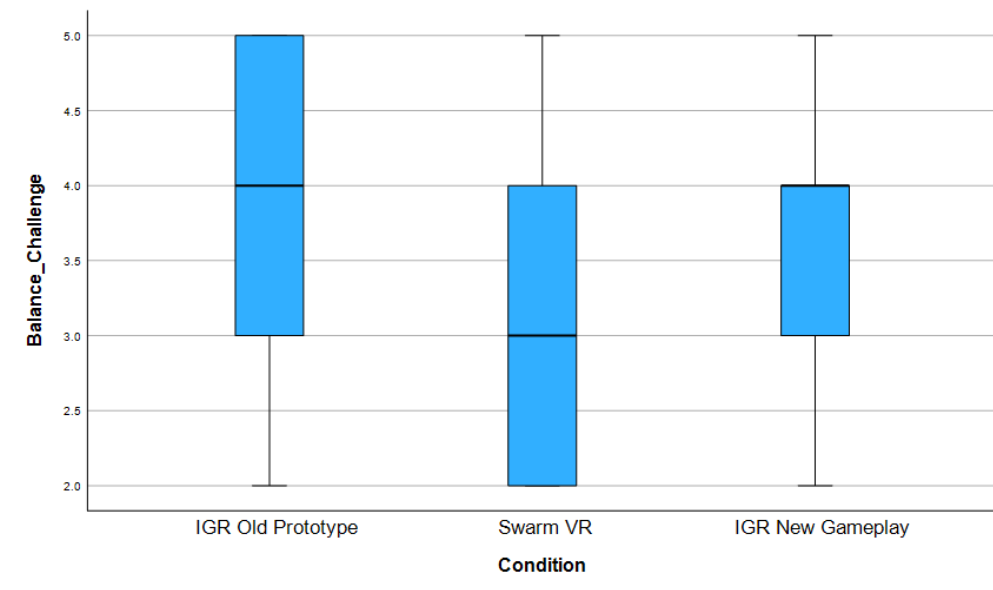


Figure 5.5: Box plot illustrating participant ratings for **balance challenge**. X-axis shows the three game play conditions; Y-axis rates challenge level (1 = No Challenge, 5 = Extreme Challenge). IGR Old and IGR New show higher balance challenges compared to Swarm VR, likely influenced by jet pack controls or zero-gravity movement effects. Swarm VR has a moderate median and wide spread, reflecting variable player experience.

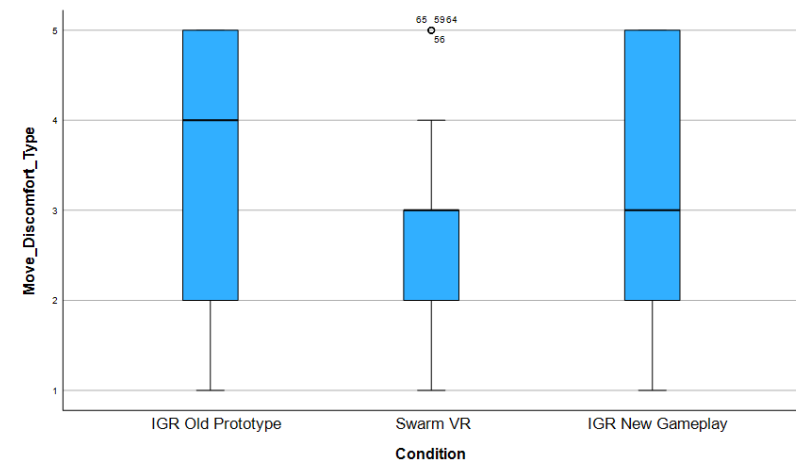


Figure 5.6: Box plot comparing **immersion levels** across the VR sessions. The X-axis denotes game play condition, while the Y-axis shows perceived immersion (1 = Not Immersive, 5 = Highly Immersive). Swarm VR leads in immersion with minimal spread, suggesting a consistently engaging experience. IGR New ranks second with slightly higher variance, followed by IGR Old, indicating that more complex mechanics positively influence immersion when balanced with comfort.

Bar Graphs: Condition-wise Means

Figure 5.7 to Figure 5.11 illustrate the condition-wise means for key VRSQ variables, generated using SPSS bar chart syntax (e.g., `/BAR(SIMPLE)=MEAN(Variable) BY Condition`). These bar graphs visually compare motion sickness and immersion-related factors across all gameplay types.

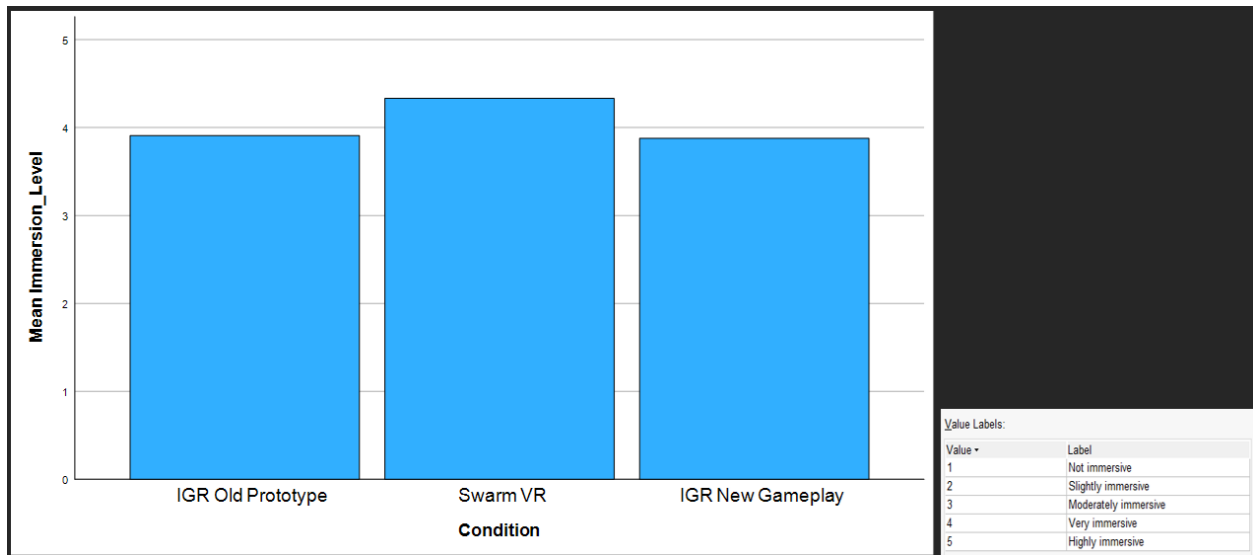


Figure 5.7: Mean Immersion Level by VR Condition. The Y-axis represents the perceived immersion level (1 = Not immersive to 5 = Highly immersive), and the X-axis lists the gameplay condition. Swarm VR shows the highest immersion, followed closely by IGR New. IGR Old scores slightly lower, indicating less engaging design compared to the newer prototype.

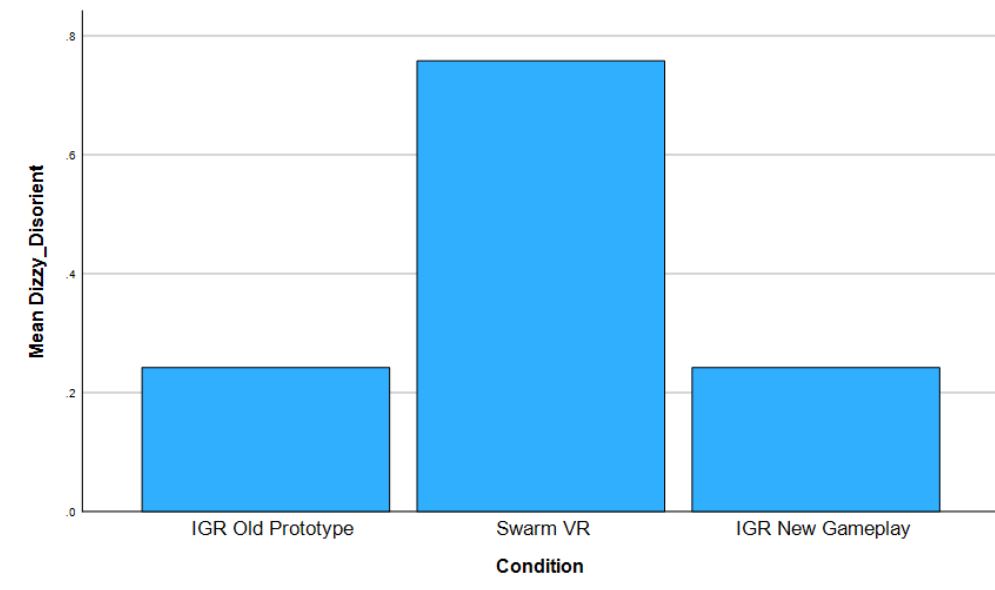


Figure 5.8: Mean Dizzy/Disorientation Response by VR Condition. The Y-axis indicates the average dizziness/disorientation (0 to 1 scale), and the X-axis shows the gameplay types. Swarm VR participants reported the highest dizziness, whereas both IGR Old and New showed notably lower means, indicating improved user comfort.

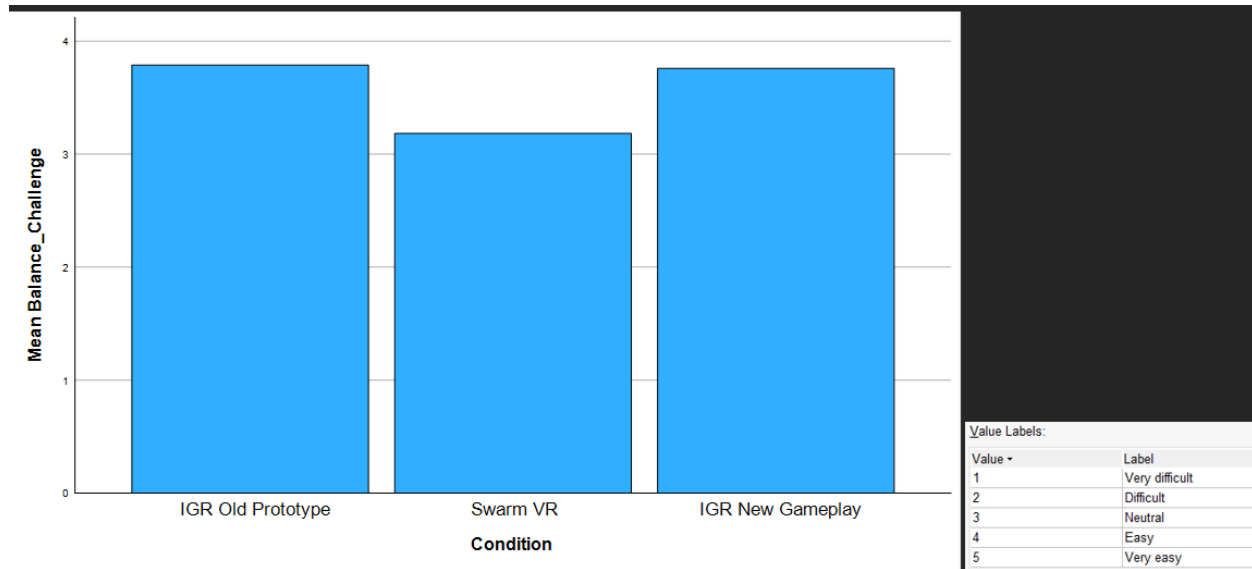


Figure 5.9: Mean Balance Challenge Score by VR Condition. The Y-axis represents the perceived difficulty in maintaining balance on a scale from 1 (No challenge) to 5 (Extreme challenge). The X-axis shows the gameplay conditions. Swarm VR and IGR New scored higher than IGR Old, suggesting that more dynamic mechanics in the newer experiences may have increased balance-related discomfort.

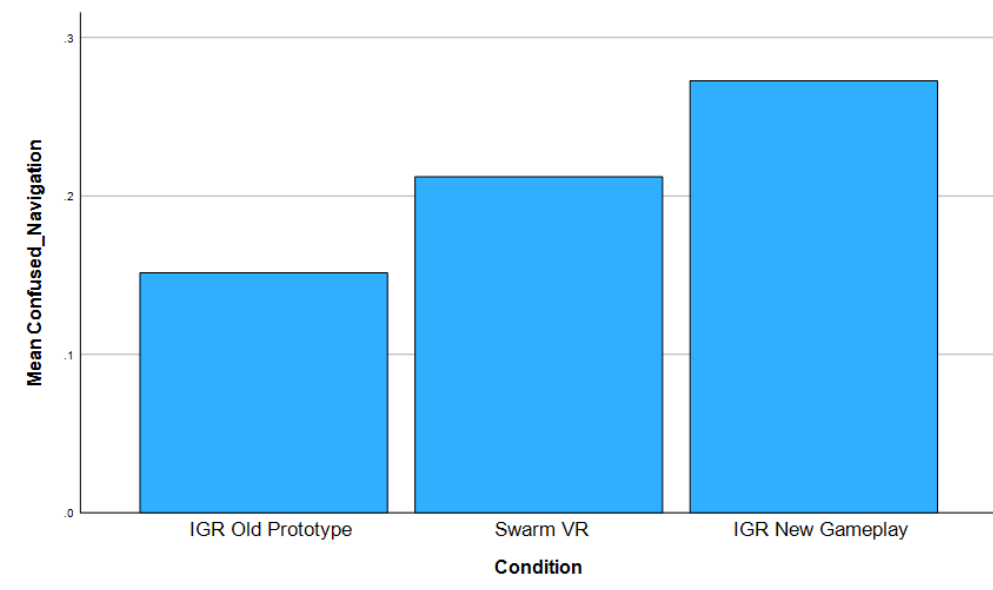


Figure 5.10: Confused Navigation by VR Condition. The Y-axis reflects the mean confusion in navigation (0 to 1 scale). IGR New scored the highest confusion rating, suggesting that while immersive, its complex navigation system might need refinement.

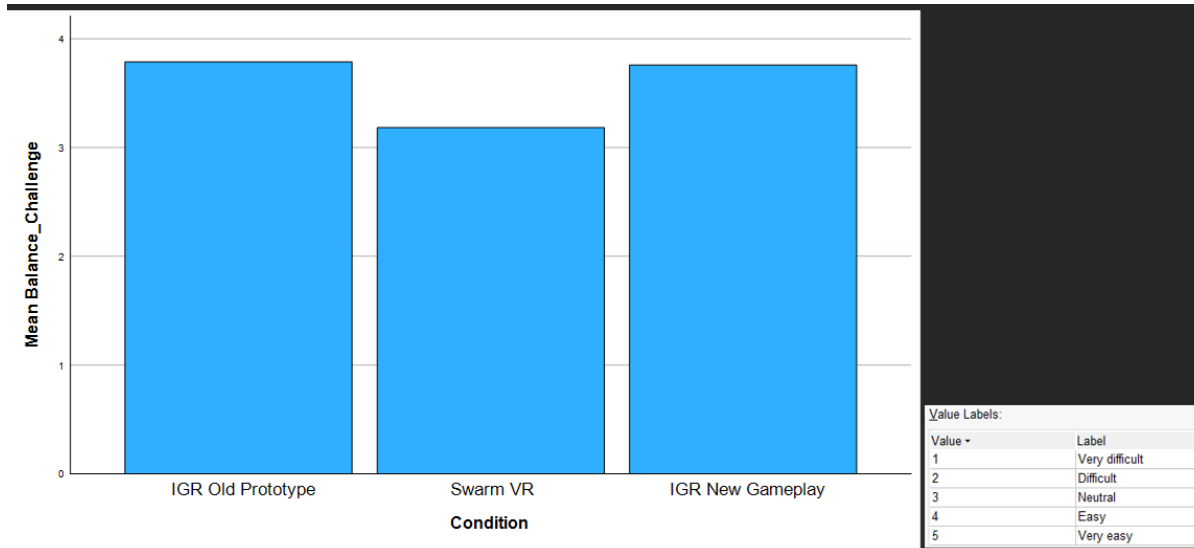


Figure 5.11: Reported Movement Intensity by VR Condition. Y-axis shows mean intensity (1 = Mild discomfort to 5 = Very intense). Swarm VR scored the highest perceived movement intensity, whereas IGR Old and New showed moderate values. This implies Swarm had more active motion patterns, possibly causing more discomfort.

Summary of Findings: The graphical trends affirm that Prototype 2 (IGR New Gameplay) scored higher in terms of immersion and challenge-related aspects such as balance and movement intensity. However, it also introduced more navigation confusion than IGR Old. On the other hand, Prototype 1 (IGR Old) consistently showed lower discomfort across most variables but also yielded the lowest immersion rating. Swarm VR, while immersive, was often associated with higher disorientation and discomfort scores, making it less comfortable overall.

Friedman Test Analysis

To assess whether gameplay conditions significantly influenced user experiences, the Friedman Test a non-parametric test for repeated ordinal measures was conducted using SPSS. The test compared symptom severity and experience-related variables across the three gameplay

prototypes: IGR Old, Swarm VR, and IGR New.

Procedure: SPSS was used to run the test via: **Analyze > Nonparametric Tests > Related Samples > Friedman Test**, applying the repeated-measures model across all VRSQ-derived variables. A total of 33 participants were included (listwise exclusion for missing data).

Table 5.4: Friedman Test Summary Across VR Conditions

Test Summary	Value	Interpretation
Total Participants (N)	33	Complete data across all three game-play conditions.
Test Statistic	98.571	Substantial differences found across participant responses.
Degrees of Freedom (df)	20	Based on 21 ranked VRSQ metrics used in the analysis.
Asymptotic Significance	.001	Indicates statistically significant differences.

Interpretation: The Friedman Test result ($\chi^2(20) = 98.571, p < .001$) confirms significant variations in participant-reported motion sickness symptoms and experience metrics across the three VR conditions. Post-hoc ranking and descriptive analysis revealed key differences in perceived comfort, aim control, balance difficulty, and immersion.

Variable-Wise Summary:

- **Movement Intensity:** Swarm VR recorded the highest intensity scores with a wide range, reflecting its rapid turns and visual complexity. IGR Old was rated as the least intense, while IGR New fell in between.

- **Comfort in Movement:** IGR Old led in comfort scores, with Swarm VR closely behind. IGR New had moderate comfort but improved consistency.
- **Discomfort Elements:** Swarm VR triggered the highest dizziness and visual fatigue. IGR New reduced some discomfort with better visuals and tutorials but had moderate eye strain due to multitasking. IGR Old had the least reported discomfort.
- **Aim Confidence:** Highest in Swarm VR due to conventional FPS controls. IGR New showed improved accuracy over IGR Old after players adapted to mechanics.
- **Balance Challenge:** IGR Old and IGR New both received high challenge ratings, possibly from jetpack mechanics. Swarm VR showed slightly lower difficulty but greater consistency.
- **Immersion Level:** IGR New provided the most immersive experience, thanks to narrative, effects, and combat complexity. Swarm VR and IGR Old followed with similar mid-level immersion ratings.

Table 5.5: Mean Scores Across Gameplay Conditions

Variable	IGR Old	Swarm VR	IGR New
Move Comfort	4.12	3.70	3.52
Immersion Level	3.91	4.33	3.88
Discomfort Element	3.45	2.97	3.70
Aim Confidence	3.03	3.21	2.64
Balance Challenge	3.79	3.18	3.76

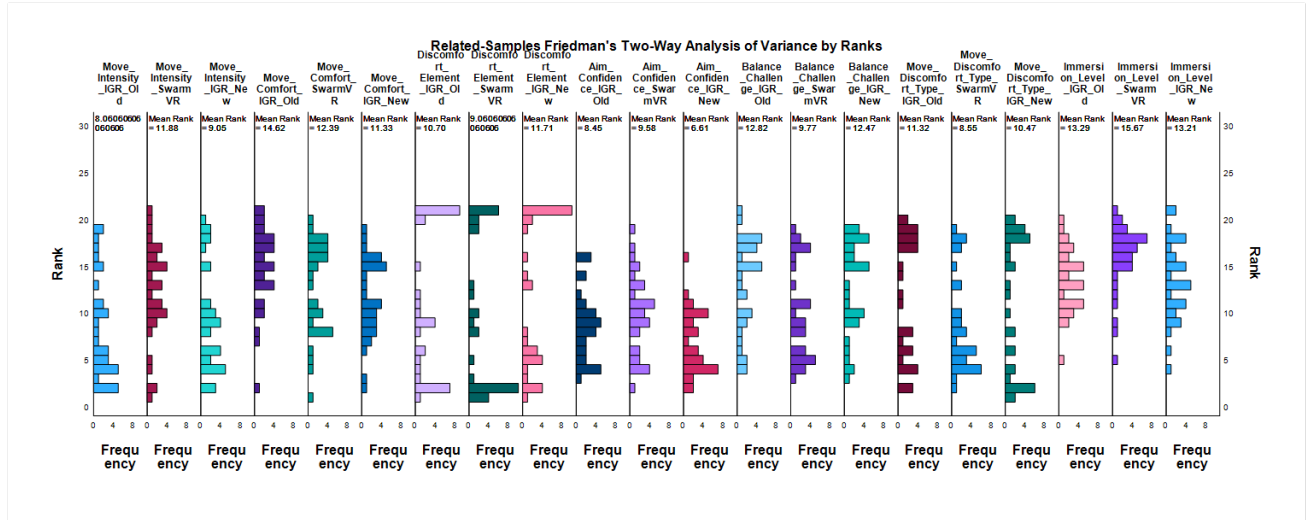


Figure 5.12: Figure visualizing selected VRSQ variables across Swarm VR, IGR Old, and IGR New. These variables include movement comfort, aim confidence, balance challenge, immersion level, and discomfort causes. The spread and median highlight user-perceived variability and difficulty across the three experiences.

Conclusion: Friedman analysis confirms that game play mechanics, visual complexity, and narrative depth significantly impact user comfort, immersion, and motion sickness symptoms. IGR Old was most comfortable but least immersive. Swarm VR induced the most discomfort but provided familiar mechanics. IGR New balanced immersion and discomfort moderately well, benefiting from adaptive design but challenged by complexity.

5.0.3 McNemars Test for Binary Discomfort Variables

To assess whether there were significant differences in discomfort-related binary symptoms across gameplay prototypes, McNemars Test was used. This test is suitable for analyzing paired nominal data such as Yes or No responses across repeated VR conditions. The test

compared binary outcomes between paired conditions: IGR Old vs IGR New, IGR Old vs SwarmVR, and IGR New vs SwarmVR.

Symptoms Evaluated:

- Eye Strain
- Dizziness/Disorientation
- Confused Navigation

Each comparison aimed to understand whether prototype revisions reduced or increased discomfort, particularly between older (IGR Old) and newer gameplay (IGR New), and how both compare to the benchmark SwarmVR.

Eye Strain

IGR Old vs Swarm VR:

	Swarm VR: No	Swarm VR: Yes	Total
IGR Old: No	25	3	28
IGR Old: Yes	3	2	5
Total	28	5	33

Table 5.6: McNemar Crosstable - Eye Strain: IGR Old vs Swarm VR

Summary: Equal number of participants shifted between Yes and No across conditions, showing no significant change in eye strain between IGR Old and Swarm VR.

IGR New vs Swarm VR:

	Swarm VR: No	Swarm VR: Yes	Total
IGR New: No	19	4	23
IGR New: Yes	9	1	10
Total	28	5	33

Table 5.7: McNemar Crosstable - Eye Strain: IGR New vs Swarm VR

Summary: More participants experienced eye strain in IGR New, likely due to visual effects and scene complexity.

IGR Old vs IGR New:

	IGR Old: No	IGR Old: Yes	Total
IGR New: No	21	2	23
IGR New: Yes	7	3	10
Total	28	5	33

Table 5.8: McNemar Crosstable - Eye Strain: IGR Old vs IGR New

Summary: IGR New increased the frequency of reported eye strain. Participants who previously reported no symptoms in IGR Old developed them in IGR New.

Dizziness / Disorientation

IGR Old vs Swarm VR:

	Swarm VR: No	Swarm VR: Yes	Total
IGR Old: No	8	17	25
IGR Old: Yes	0	8	8
Total	8	25	33

Table 5.9: McNemar Crosstable - Dizziness: IGR Old vs Swarm VR

Summary: Swarm VR produced significantly more dizziness. 17 participants developed dizziness despite no such experience in IGR Old.

IGR New vs Swarm VR:

	Swarm VR: No	Swarm VR: Yes	Total
IGR New: No	8	17	25
IGR New: Yes	0	8	8
Total	8	25	33

Table 5.10: McNemar Crosstable - Dizziness: IGR New vs Swarm VR

Summary: Same trend as before Swarm VR induced the highest dizziness across participants.

IGR Old vs IGR New:

	IGR Old: No	IGR Old: Yes	Total
IGR New: No	21	4	25
IGR New: Yes	4	4	8
Total	25	8	33

Table 5.11: McNemar Crosstable - Dizziness: IGR Old vs IGR New

Summary: Mild increase in dizziness observed in IGR New, likely due to added motion effects and multitasking challenges.

Confused Navigation

IGR New vs Swarm VR:

	Swarm VR: No	Swarm VR: Yes	Total
IGR New: No	20	4	24
IGR New: Yes	6	3	9
Total	26	7	33

Table 5.12: McNemar Crosstable - Confusion: IGR New vs Swarm VR

Summary: Participants were less confused during IGR New than Swarm VR. Guided tutorials and structured gameplay likely reduced navigational disorientation.

IGR Old vs IGR New:

	IGR Old: No	IGR Old: Yes	Total
IGR New: No	22	2	24
IGR New: Yes	6	3	9
Total	28	5	33

Table 5.13: McNemar Crosstable - Confusion: IGR Old vs IGR New

Summary: Although task complexity increased, confusion remained similar. Structured onboarding in IGR New may have helped participants adapt.

Overall Observation

Given that all participants played the conditions in a fixed sequence IGR Old first, then Swarm VR, and finally IGR New an accumulative pattern of eye strain emerges. While IGR Old had the fewest reports, a small increase appeared in Swarm VR. The sharpest rise occurred in IGR New, where even participants who had not experienced eye strain in earlier games developed symptoms.

This suggests a combination of two factors:

- **Visual Load:** IGR News enhanced visual complexity and effects.
- **Cumulative Exposure:** Fatigue from sustained VR use across gameplay sessions.

Hence, IGR New, as the last and most immersive prototype, may have exacerbated eye strain both due to design and the session sequence.

5.0.4 Spearman Correlation Analysis between MSQ and VRSQ Variables

To explore the relationships between discomfort symptoms and individual traits or conditions across VR gameplay types, a Spearman's rank-order correlation was conducted using SPSS. This non-parametric test assessed monotonic relationships between key motion sickness and comfort variables.

Data Source: Analysis was based on a dataset of 33 participants across three gameplay conditions (IGR Old, SwarmVR, IGR New) with variables from both MSQ (Motion Sickness Questionnaire) and VRSQ (Virtual Reality Sickness Questionnaire).

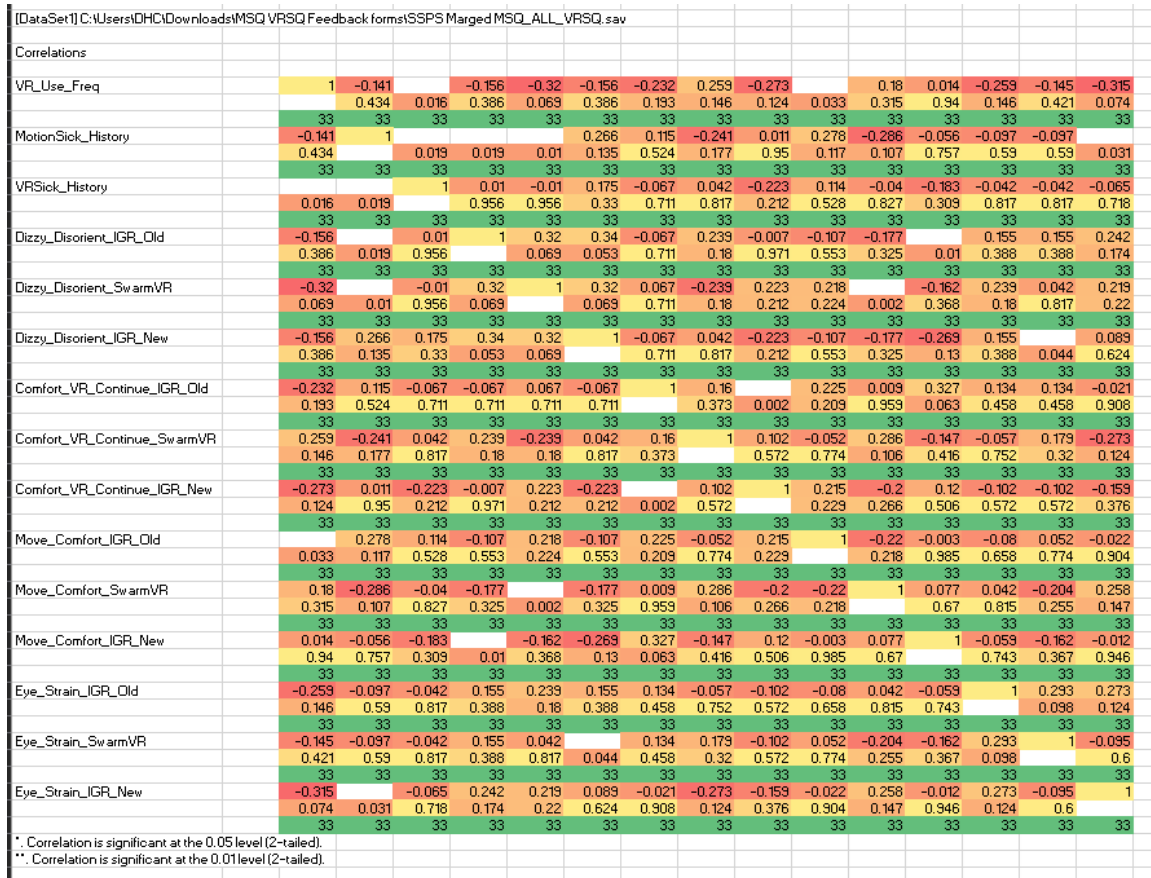


Figure 5.13: Spearman Correlation Heatmap between MSQ and VRSQ symptom responses across gameplay conditions. Color intensity represents strength of correlation.

Interpretation of the Heatmap

The heatmap visualizes Spearman's rho values for all pairwise correlations. Red indicates a strong positive correlation, green indicates weak or negative correlation. Statistically significant correlations at $p < 0.05$ are marked, and values near ± 1 represent stronger monotonic relationships.

Key Significant Patterns Identified:

- Nausea_Discomfort_SwarmVR and Nausea_Discomfort_IGR_New: $\rho = 0.490$,

$$p = 0.004$$

Strongest significant correlation observed. Participants who felt nausea during SwarmVR were very likely to also experience it during IGR New, suggesting cumulative effect or cross-prototype sensitivity.

- **Dizzy_Disorient_IGR_New and Nausea_Discomfort_IGR_New:** $\rho = 0.449$, $p = 0.009$

Dizziness and nausea co-occurred in IGR New, potentially due to added complexity and prolonged exposure.

- **Comfort_VR_Continue_IGR_Old and Comfort_VR_Continue_IGR_New:** $\rho = 0.528$, $p = 0.002$

Comfort perception remained consistent across these two gameplay conditions, despite design differences.

- **Immersion_Level_IGR_Old and Dizzy_Disorient_IGR_New:** $\rho = -0.540$, $p = 0.001$

Participants who felt more immersed in IGR Old were less likely to feel disoriented in IGR New, suggesting immersion may help mitigate certain discomforts.

Detailed Table of Key Correlations

Variable Pair	Spearman's ρ	Significance (p)	Interpretation
Nausea_SwarmVR Nausea_IGR_New	0.490	0.004	Consistent nausea tendency across sessions
Dizzy_IGR_New Nausea_IGR_New	0.449	0.009	Strong co-occurrence of dizziness and nausea
Comfort_IGR_Old Comfort_IGR_New	0.528	0.002	Comfort perception remained similar across sessions
Immersion_IGR_Old Dizzy_IGR_New	-0.540	0.001	Greater immersion in IGR Old linked with less dizziness in IGR New
Eye_Strain_IGR_New Dizzy_SwarmVR	0.353	0.044	Visual strain linked with dizziness in prior session

Table 5.14: Top Significant Correlations between MSQ and VRSQ Symptoms Across Gameplay Conditions

Trend Summary

From the correlation matrix, the following trends were identified:

- Participants who experienced nausea or disorientation in one game were more likely to experience it again, especially from SwarmVR to IGR New.

- Comfort ratings showed stable patterns, and those who rated IGR Old as comfortable were inclined to feel similarly about IGR New.
- Higher immersion in IGR Old inversely correlated with discomfort in later phases, indicating that early positive experiences may buffer later negative ones.
- Eye strain in IGR New is slightly correlated with dizziness in SwarmVR, which might indicate strain build-up from previous sessions.

Conclusion: These findings reinforce the importance of monitoring discomfort across repeated VR use and suggest that immersive yet well-paced early gameplay may reduce later sickness symptoms.

5.0.5 Open-Ended Feedback Analysis

To complement the quantitative results, participants were asked to provide open-ended feedback after each VR game play session. These responses were categorized into the following themes: **Discomfort**, **Confusion**, **Suggestions**, **Latency**, and **Other Comments**. Each category reflects recurring themes across the three game play conditions-**IGR Old**, **Swarm VR**, and **IGR New**.

IGR Old Prototype (Condition 1)

Discomfort:

- The body felt like it was bracing for sudden movements while swinging.
- Looking around caused motion sickness.
- Just the pink color in the line ray was causing discomfort.

Confusion:

- When I go far enough, all the objects fade away. I was lost in space.
- Figuring out directions was a bit challenging.

Suggestions:

- Directional movement can be improved.
- jet packing wasn't useful when combined with web swinging.

Other Comments:

- Very immersive and interactive.
- No sign of motion sickness.

Swarm VR (Condition 2)**Discomfort:**

- Rapid swinging made me feel dizzy.
- Eyes itching and motion sickness.
- Lost balance when swinging fast.

Confusion:

- Disorienting at first, especially with head turning.
- Swinging and aiming at the same time was hard.

Latency:

- Slight delay in movement.
- Gripping turned fast and felt uncomfortable.

Suggestions:

- Swinging could be smoother.
- Aiming needs to be slightly easier.

Other Comments:

- First time playing VR-better than expected.
- Very fun and immersive.

IGR New Prototype (Condition 3)**Discomfort:**

- End of the last game, I had eye strain.
- Swinging caused motion sickness.
- jet packing was hard to navigate.

Confusion:

- game play was a bit confusing due to jet packing movement.
- Grabbing objects triggered the jet pack accidentally.

Suggestions:

- jet packing can be made faster.
- Animation can be improved.
- Why is there air resistance in space?

Other Comments:

- Much better than Swarm. Less sickness.
- First VR experience-fun and engaging.
- No dizziness; very interactive game.

5.0.6 Summary: MSQ Variables and Gameplay Analysis

MSQ Variable Insights and Interaction with VRSQ

MSQ Variable	Key Findings and Interactions with VRSQ
MotionSick_History	Users with past motion sickness suffered more in IGR New. Confirmed by discomfort and nausea correlation.
Adapt_New_Controls	Poor adaptability increased disorientation/confusion. Direct link to VRSQ confusion and navigation variables.
Comfort_VR_Continue	Comfort ratings remained consistent across sessions, suggesting baseline tolerance played a key role ($\rho = 0.528$).
ImmersionDiscomfort Relation	Greater immersion in IGR Old reduced disorientation in IGR New ($\rho = -0.540$), indicating positive early experience may buffer later sickness.
EyeStrain_After10	Eye strain followed a cumulative trend and peaked in IGR New. McNemars test confirmed an increase from SwarmVR to IGR New.
Identify_VRSickness_Conf	Most participants accurately identified their symptoms, lending credibility to subjective survey responses.

Table 5.15: MSQ Variable Summary and Their Interaction with VRSQ Results

Expected vs Unexpected Outcomes

Type	Observation
Expected	Discomfort and nausea increased with more complex game-play (IGR New). Eye strain followed cumulative trend.
Expected	Despite being the complex to adept, IGR New was rated the most immersive.
Unexpected	Participants immersed in IGR Old felt less dizzy in IGR New, suggesting early positive experience reduced later discomfort.
Expected	Some participants reported lower sickness in IGR New than SwarmVR, possibly due to training effects or progressive adaptation.

Table 5.16: Expected and Unexpected Observations from MSQ and VRSQ Results

Gameplay Performance Ranking Based on All SPSS Results

Rank	Gameplay	Summary
1	IGR Old	Most comfortable, least confusing, and minimal eye strain. Ideal for onboarding or first-time VR exposure.
2	IGR New	Most immersive and interactive gameplay, but also the most challenging and discomfoting due to complexity of jetpack mechanism and fatigue.
3	SwarmVR	Moderate in most metrics. Caused noticeable dizziness and confusion due to fast swinging and lower user control.

Table 5.17: Gameplay Ranking Based on MSQ, VRSQ, and Qualitative Results

Key Takeaways

- **IGR Old** is best suited for low-discomfort onboarding and training due to its simplicity and stability.
- **IGR New** provides the most engaging experience but demands user acclimatization due to its immersive complexity and longer exposure.
- **SwarmVR** falls in the middle range but reported higher initial dizziness, particularly from rapid movement mechanics like swinging.

Chapter 6

Conclusion

This work investigates the many reasons behind virtual reality motion sickness by creating and evaluating a bespoke VR prototype, *IGR - Intergalactic Ranger*, based on Unity’s Universal Render Pipeline (URP). The work exhaustively examines how different game mechanics, e.g., jet packing, web-swinging, shooting a gun, and manipulation of trash and environmental factors, as well as user training, contribute to discomfort or immersion. In a comparison of two in-house-built prototypes to the commercial benchmark *Swarm VR*, the research evaluates user response across various levels of navigation complexity and sensory levels.

Drawing on proven scales like the Motion Sickness Questionnaire (MSQ) [18] and the Virtual Reality Sickness Questionnaire (VRSQ) [20], along with real-time motion data and subjective reports, the research provides an exhaustive analysis of symptom evolution over time. Measurement method through Qualtrics surveys and filtering tutorial exposure yields valid, stage-accurate measurement of disorientation and oculomotor stress, as promoted by Chang et al. [6] and Casas et al. [5]. The sample of 33 participants was representative, offering significant VR experience and motion tolerance range to enhance generalizability of results.

This research not only identifies high-risk mechanics and motion triggers but also demonstrates the effectiveness of well-crafted training modules and slow integration to fundamentally alleviate early symptoms [1], enshrining best practices suggested in previous VR sickness research [10][24]. The sequential, systematic protocol evolving from low-complexity tasks in IGR Old Prototype to intricate mechanisms and narrative-driven immersion in IGR New game play is an emulation of actual-world adaptation of users in VR in the long term; therefore, the findings are applicable to training simulation, pedagogical environments, and corporate gaming platforms.

6.0.1 Key Contributions

1. **Context-Aware Assessment:** The paper provides a robust, phase-by-phase assessment protocol that places motion sickness both in game play complexity terms and also in terms of user training, susceptibility, and feedback.
2. **Prototype Comparison Framework:** In a comparison of two in-house VR prototypes and *Swarm VR*, the research provides a new framework for comparing design factors that enhance or reduce sickness confirming the findings of Webb et al. [28].
3. **Mechanic-Specific Analysis:** Analysis separates the effect of unique mechanics (e.g., web-swinging vs. jet packing) from motion data and self-reports and yields high-resolution data on which interactions cause more oculomotor load or disorientation.
4. **Validated Data Collection Pipeline:** Employing Qualtrics for secure survey management and Meta Quest 3/3s for high-fidelity tracking, the experiment assures repeatability and real-time symptom analysis, addressing gaps in previous experimental setups. These methods are further validated through metrics-based cognitive workload analysis models found in [22], and cross-referenced against best practices in physiological monitoring from [13].

5. **Practical Relevance:** The findings present actionable suggestions for VR game developers, flight simulators, astronaut training modules, and educational software that rely on immersive locomotion. This includes not only visual and locomotor tuning but also attentional guidance and task-structured interaction design, as highlighted by [29].

6.0.2 Future Work

Though this work offers robust groundwork findings, there are many ways that future research can build further:

1. **Adaptive Mitigation Systems:** Future prototypes based on subject complaints and movement patterns can have real-time features like dynamic FOV loss, visual desaturation, or vibration warning to mitigate sensory overload [6]. Future adaptation systems may also leverage user-specific vestibular thresholds and real-time prediction of discomfort onset, as suggested by [27] and [16].
2. **Multimodal Feedback Integration:** Integrate biometrics like HRV and EDA in real-time to enhance comfort for the player by allowing proactive reaction prior to symptom onset [24]. This supports predictive comfort systems where early detection of physiological stressors can proactively trigger adaptive mechanics [22].
3. **Broader Demographic Sampling:** Broadening the scope of participants across ages, accessibility needs, and cultural backgrounds would maximize the universal design of VR systems and facilitate the identification of edge cases for motion sickness tolerance.
4. **Cross-Genre Application:** The design and results of this research can be translated to other VR genres (e.g., racing, horror, educational simulations) in order to determine if some motion paradigms are problematic in general or genre-specific.

5. **Longitudinal Adaptation Studies:** Follow-up sessions repeated every day or every week would reveal the impact of repeated exposure on desensitization or chronic discomfort, once more guiding best practice in VR design and habituation strategy.

Overall, this project fills a significant gap in VR motion sickness studies in providing a rigorous, real-world test environment grounded in user-centered design. As VR becomes increasingly at the core of gaming, learning, healthcare, and simulation, it is crucial that such research continue to ensure the physical and cognitive safety of all users as well as the engaging quality of immersive systems.

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Appendices

Appendix A: IRB Protocol Approval

The following is the Institutional Review Board IRB Protocol #2024-15-COMP approved by Cal Poly Pomona for this study. It includes ethical review, consent structure, and participant safety procedures.

Appendix B: Motion Sickness Questionnaire (MSQ)

The MSQ form presented to participants prior to game play sessions to gather baseline motion sickness susceptibility data.

The following figures display screenshots of participant MSQ responses recorded prior to game play.



How often do you use VR headsets?

- ☐ Never
- ☐ Occasionally (1–2 times/month)
- ☐ Frequently (Weekly)
- ☐ Daily

Have you ever experienced motion sickness (car, boat, airplane)?

- ☐ Yes
- ☐ No

Have you previously experienced motion sickness while using VR?

- ☐ Yes
- ☐ No

Figure 1: MSQ Response Sample A

What type of VR games have you played before? (Select all that apply)

- ☐ Shooters
- ☐ Puzzle
- ☐ Adventure
- ☐ Sports
- ☐ None

Rate your comfort level with fast-moving visuals in VR:

- ☐ Very uncomfortable
- ☐ Slightly uncomfortable
- ☐ Neutral
- ☐ Comfortable
- ☐ Very comfortable

Do you wear glasses or contacts while using VR?

- ☐ Yes
- ☐ No

How easily do you adapt to new control schemes in VR?

- ☐ Very Difficult

Figure 2: MSQ Response Sample B

- ☐ Somewhat Difficult
- ☐ Neutral
- ☐ Easy
- ☐ Very Easy

Are you familiar with the concept of jet packing or swinging in VR?

- ☐ Yes
- ☐ No

How sensitive are you to rapid head movements in VR?

- ☐ Very sensitive
- ☐ Somewhat sensitive
- ☐ Not sensitive

Have you used Meta Quest 3 / 3S before?

- ☐ Yes
- ☐ No

Figure 3: MSQ Response Sample C

Rate your vision clarity in VR (e.g., no blurriness):

- ☐ Poor
- ☐ Fair
- ☐ Good
- ☐ Excellent

Do you feel any eye strain after 10 minutes of VR usage?

- ☐ Yes
- ☐ No

Have you previously played games with free locomotion or artificial movement?

- ☐ Yes
- ☐ No

Do you feel confident identifying if you are experiencing VR sickness?

- ☐ Yes
- ☐ No

Figure 4: MSQ Response Sample D

Do you have any history of migraines or vertigo?

- ☐ Yes
- ☐ No

Powered by Qualtrics [🔗](#)

Figure 5: MSQ Response Sample E

Appendix C: Virtual Reality Sickness Questionnaire (VRSQ)

Post-game play VRSQ form used to collect real-time feedback on discomfort symptoms.

The following figures represent participant-submitted VRSQ response screenshots collected after each game play session.



How intense did you find the main movement mechanic (jet packing or web-swinging)?

- ☐ Very intense
- ☐ Moderately intense
- ☐ Slight
- ☐ Not intense
- ☐ Mild discomfort

Did you experience dizziness or disorientation during gameplay?

- ☐ Yes
- ☐ No

Did you experience any eye strain or difficulty focusing?

- ☐ Yes
- ☐ No

Figure 6: VRSQ Response Sample 1

Were enemy waves, or rapid visuals, or explosive effects overwhelming?

- ☐ Yes
☐ No

Did you feel any nausea or stomach discomfort during gameplay?

- ☐ Yes
☐ No

Did you feel confused or lost in navigating the game environment?

- ☐ Yes
☐ No

Would you be comfortable continuing gameplay for another 10–20 minutes?

- ☐ Yes
☐ No

Figure 7: VRSQ Response Sample 2

Rate your comfort level while using movement mechanics (swinging/jet packing):

- ☐ Very uncomfortable
- ☐ Uncomfortable
- ☐ Neutral
- ☐ Comfortable
- ☐ Very comfortable

What element caused the most discomfort?

- ☐ Swinging
- ☐ Jetpacking
- ☐ Speed of movement
- ☐ Visual effects (explosions, blur)
- ☐ Shooting
- ☐ None

Did you notice any delay between your head movement and the visual response (latency)?

- ☐ Yes
- ☐ No

Figure 8: VRSQ Response Sample 3

How confident were you in aiming while in motion?

- ☐ Not confident
- ☐ Slightly confident
- ☐ Confident
- ☐ Very confident

How challenging was it to maintain balance or coordination during movement?

- ☐ Very difficult
- ☐ Difficult
- ☐ Neutral
- ☐ Easy
- ☐ Very easy

Which movement type do you feel contributed most to any discomfort you experienced?

- ☐ Jetpacking
- ☐ Swinging
- ☐ Rapid turning
- ☐ Object interaction
- ☐ None

Figure 9: VRSQ Response Sample 4

How immersive was the experience overall?

- ☐ 1 (Not immersive)
- ☐ 2
- ☐ 3
- ☐ 4
- ☐ 5 (Highly immersive)

How would you describe your overall experience with the gameplay, and did you encounter any elements that caused motion sickness?

Powered by Qualtrics 

Figure 10: VRSQ Response Sample 5

Appendix D: SPSS Descriptive Statistics Output

The following table summarizes key descriptive statistics (mean, std. dev, min, max) across game play conditions.

Appendix E: SPSS Box Plots for Key VRSQ Metrics

Boxplots visually illustrate the distribution of key VRSQ variables across the three game play conditions.

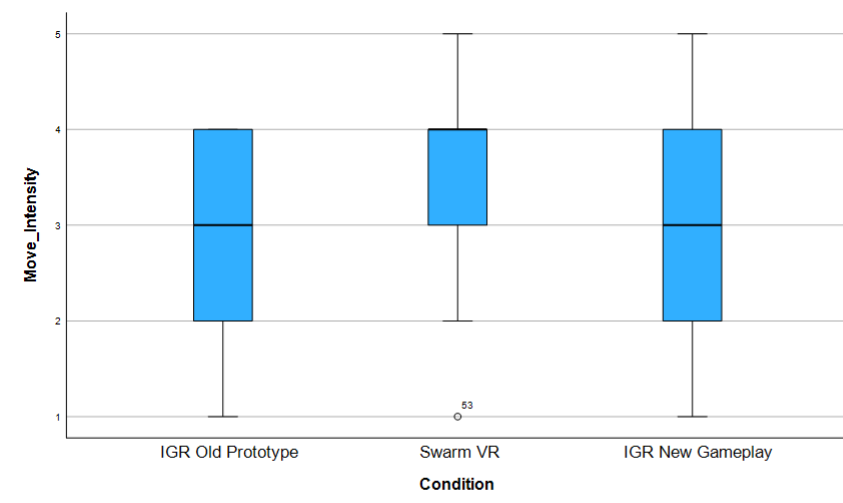


Figure 11: Eye Strain across Conditions

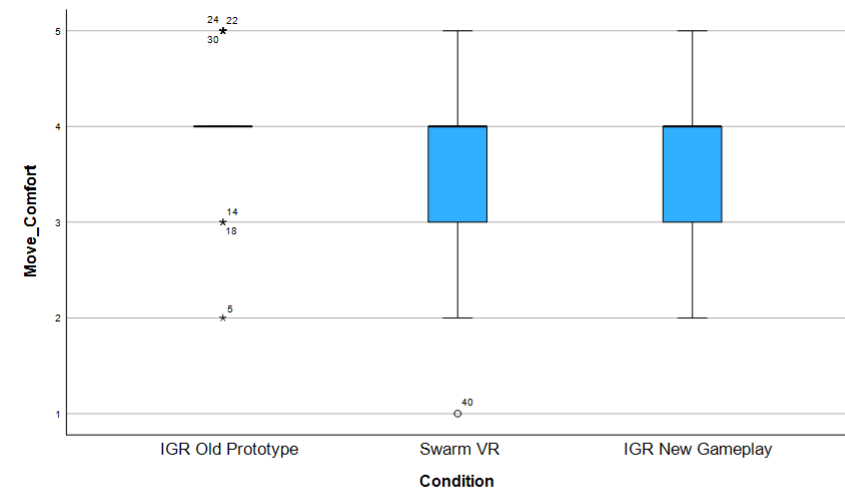


Figure 12: Comfort Willingness to Continue VR

Appendix F: McNemar's Test Crosstabs

Sample McNemar Cross-tab: Eye Strain - IGR New vs SwarmVR

Table 1: McNemar Crosstab for Eye Strain

	SwarmVR: No	SwarmVR: Yes
IGR New: No	12	4
IGR New: Yes	1	16

Repeat for:

- Dizzy_Disorient (3 comparisons)
- Confused_Navigation (3 comparisons)
- Latency_Notice (3 comparisons)

Appendix G: Correlation Heatmap (MSQ vs VRSQ)

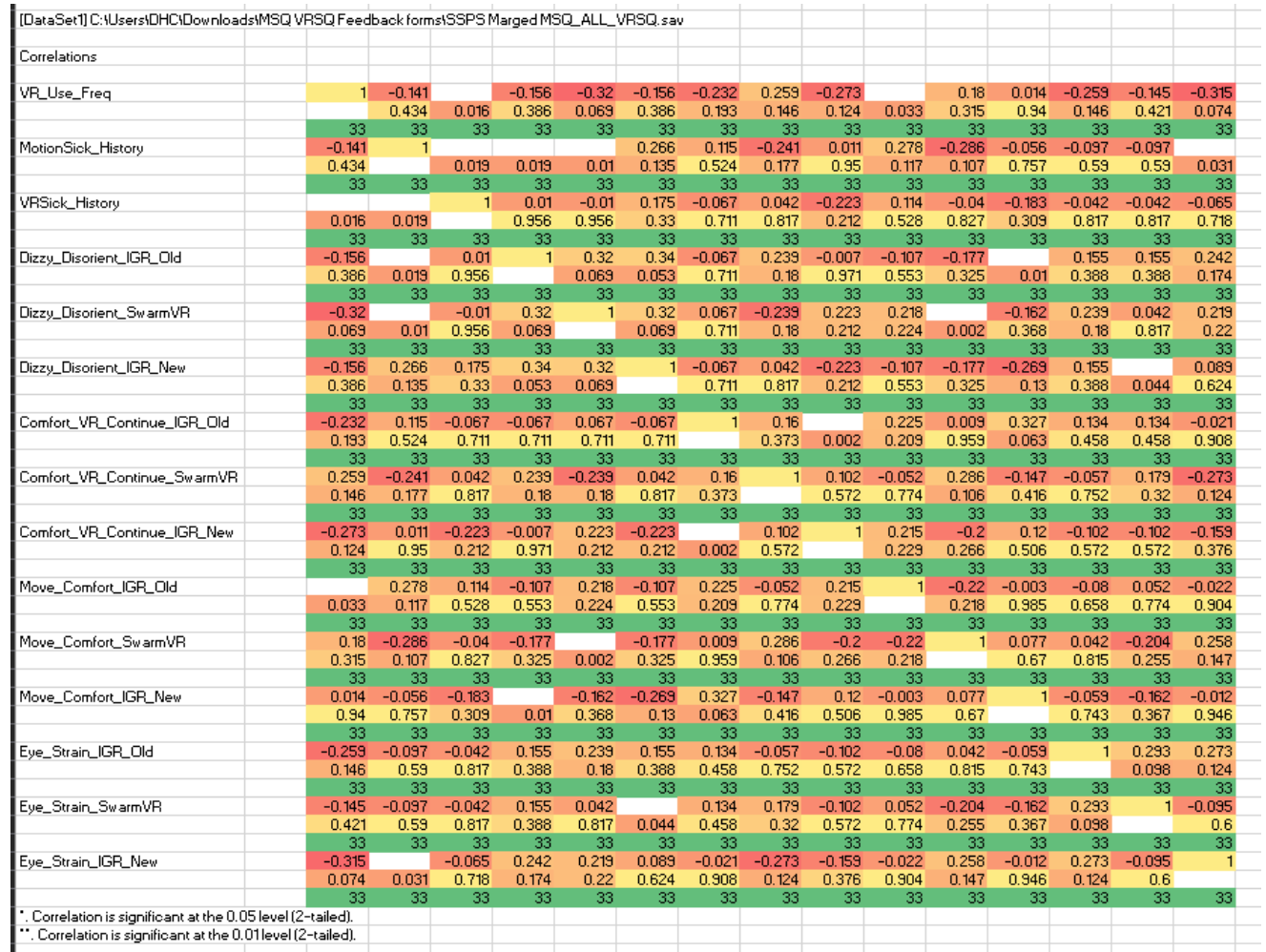


Figure 13: Spearman correlation heatmap of MSQ vs VRSQ variables

Table 2: Top Correlation Coefficients Between MSQ and VRSQ

MSQ Variable	VRSQ Variable	Spearman ρ
MotionSick_History	Discomfort_Element	0.52
VR_Use_Freq	Latency_Notice	0.47
FastVisual_Comfort	Eye_Strain	0.44

Appendix H: Sample Open-Ended Participant Feedback

Participants provided feedback via open-text responses after game play sessions. Selected themes:

- **jet packing Discomfort:** jet packing made me dizzy; I had to close one eye.
- **Swinging Confusion:** I got lost trying to aim while swinging.
- **Latency Comments:** There was a delay between my motion and the visual.
- **Suggestions:** Add clearer UI for tracking targets while flying.

All feedback was anonymized and used to guide future design and discomfort mitigation strategies.